Haloalkanes & Haloarenes

By

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जननी जन्मभूमि, स्वर्ग से महान है।
इसके वास्ते ये तन है, मन है और प्राण हैं।।

इसके कण-कण में लिखा राम-कृष्ण नाम है।
हुतात्माओं के रूधिर से भूमि शस्य-श्याम है।
धर्म का ये धाम है, सदा इसे प्रणाम है।
स्वतंत्र है धरा यहां, स्वतंत्र आसमान है।
जननी जन्मभूमि, स्वर्ग से महान है।।1।।

इसकी आन पर कभी जो बात कोई आ पड़े।
इसके सामने जो जुल्म के पहाड़ हों खड़े।।
शत्रु सब जहान हों, विरूद्ध विधि-विधान हो,
मुकाबला करेंगे जब तक जान में ये जान है।
जननी जन्मभूमि, स्वर्ग से महान है।।2।।

इसकी गोद में हजारों गंगा-यमुना झूमती।
इसके पर्वतों की चोटियां गगन को चूमती।।
भूमि ये महान है, निराली इसकी शान है।
इसकी जय-पताका ही स्वयं विजय-निशान है।
जननी जन्मभूमि, स्वर्ग से महान है।।3।।

Unit - 1: Haloalkanes **(10 Hours)**

Alkyl halides: Methods of preparation and properties, nucleophilic substitution reactions – S_N1, S_N2 and S_Ni mechanisms with stereochemical aspects and effect of solvent; nucleophilic substitution v/s elimination.

Organometallic compounds of Mg (Grignard reagent) – Use in synthesis of organic compounds.

Unit - III: Aryl halides **(04 Hours)**

Preparation (including preparation from diazonium salts) and properties, nucleophilic aromatic substitution; S_NAr, Benzyne mechanism. Relative reactivity of alkyl, allyl, benzyl, vinyl and aryl halides towards nucleophilic substitution reactions.

The bond strength of C–X decreases as the size of X increases

- The C–X bond-dissociation energies in the halomethanes, CH_3X, decrease along the series F, Cl, Br, I. At the same time, the C–X bond lengths increase (Table 6-1). The bond between carbon and halogen is made up mainly by the overlapping of an sp^3 hybrid orbital on carbon.
- with a p orbital on the halogen (Figure 6-1). In the progression from fluorine to iodine in
- the periodic table, the size of the halogen increases, and the electron cloud around the halogen atom becomes more diffuse. As a consequence of this growing mismatch in size between the halogen p orbital and the relatively compact orbital on carbon, bonding overlap diminishes along the series, leading to both longer and weaker C–X bonds. This phenomenon is general: Short bonds are stronger than longer bonds.

Table 6-1 C–X Bond Lengths and Bond Strengths in CH_3X

Halo-methane	Bond length (Å)	Bond strength [kcal mol^{-1} (kJ mol^{-1})]
CH_3F	1.385	110 (460)
CH_3Cl	1.784	85 (356)
CH_3Br	1.929	70 (293)
CH_3I	2.139	57 (238)

Increasing bond length

Decreasing bond strength

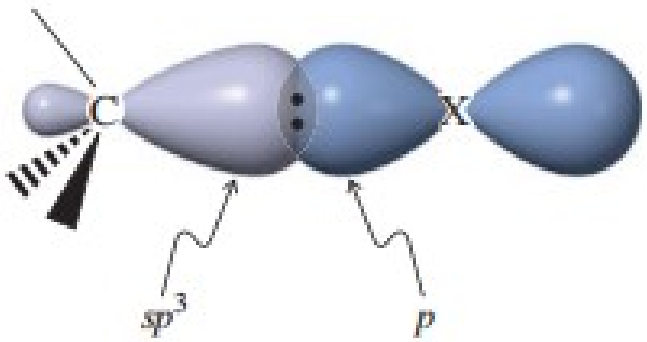


Table 6-2 Boiling Points of Haloalkanes (R–X)

	Boiling Point (°C)				
R	X = H	F	Cl	Br	I
CH_3	−161.7	−78.4	−24.2	3.6	42.4
CH_3CH_2	−88.6	−37.7	12.3	38.4	72.3
$CH_3(CH_2)_2$	−42.1	−2.5	46.6	71.0	102.5
$CH_3(CH_2)_3$	−0.5	32.5	78.4	101.6	130.5
$CH_3(CH_2)_4$	36.1	62.8	107.8	129.6	157.0
$CH_3(CH_2)_7$	125.7	142.0	182.0	200.3	225.5

Dipole–Dipole Attraction

- **Cortisol (also known as *hydrocortisone*) is a member of the class of naturally occurring compounds called *corticosteroids,* so-named because they are secreted by the cortex of the adrenal gland.**
- **The corticosteroids are anti-infl ammatory agents with numerous medicinal applications.**
- **In recent years it has become clear that infl ammation of the breathing pas_x0002_ sages is a contributing factor to the bronchoconstriction associated with attacks of asthma.**
- **Thus, oral administration of corticosteroids together with inhalation of bronchodilators has been an effective course of treatment.**
- **However, ingestion of steroids exposes patients to potential side effects.**
- **Unlike the steroids used for illicitly enhancing athletic performance corticosteroids administered at high doses can inhibit growth, a side effect that is of particular concern with children.**
- **As a result, synthetic corticosteroids have been developed that are suitable for use via inhalation and are not absorbed systemically to a signifi cant extent.**
- **One such synthetic steroid is pictured above: fluticasone propionate. The presence of haloalkane functionality has a marked effect on anti-infl ammatory potency. Fluticasone is one of a growing number of synthetic pharmaceuticals in which replacement of hydrogen by fluorine either generates novel therapeutic behavior or enhances properties already present.**

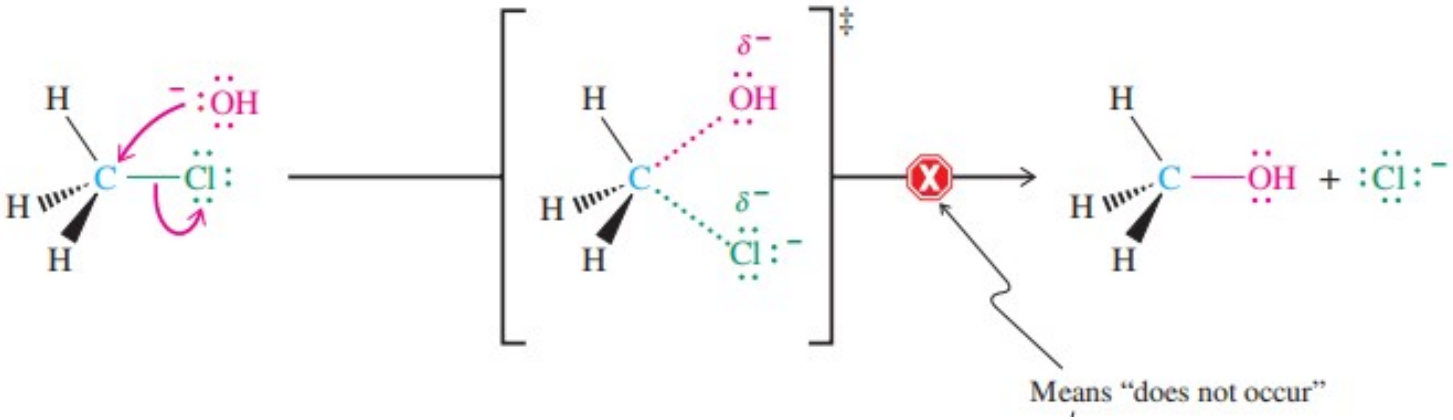
Nucleophilic Substitution

Haloalkanes contain an electrophilic carbon atom, which may react with nucleophiles substances that contain an unshared electron pair. The nucleophile can be an anion, such as hydroxide or a neutral species, such as ammonia (:NH3). In this process, which we call nucleophilic substitution, the reagent attacks the haloalkane and replaces the halide. A great many species are transformed in this way, particularly in solution. The reaction occurs widely in nature and can be controlled effectively even on an industrial scale.

The reaction of chloromethane with sodium hydroxide is bimolecular

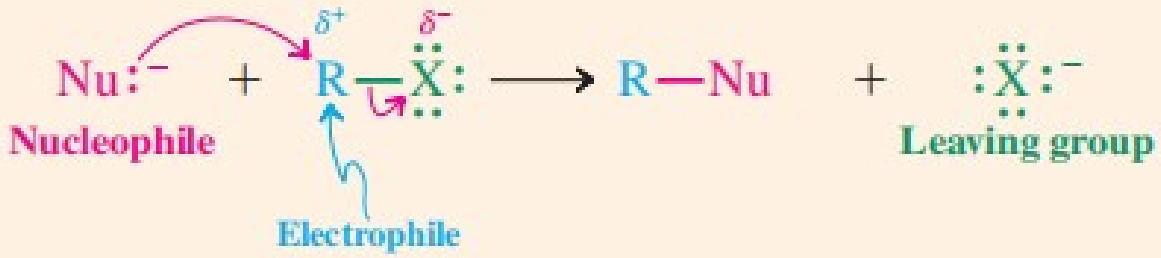
We can monitor rates by measuring either the disappearance of one of the reactants or the appearance of one of the products. When we apply this method to the reaction of chloro_x0002_
methane with sodium hydroxide, increases the rate by a factor of 4. These results are consistent with a *second-order* process which is governed by the following rate equation:

$$\text{Rate} = k[CH_3Cl][HO^-] \text{ mol L}^{-1} \text{ s}^{-1}$$

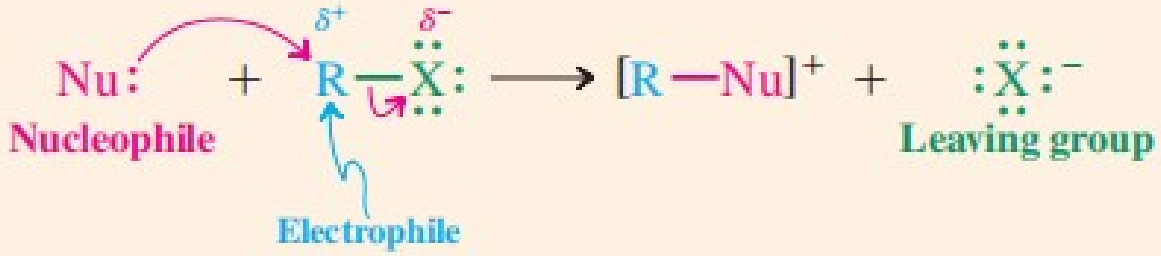


Nucleophilic Substitutions

Negative nucleophile gives neutral product



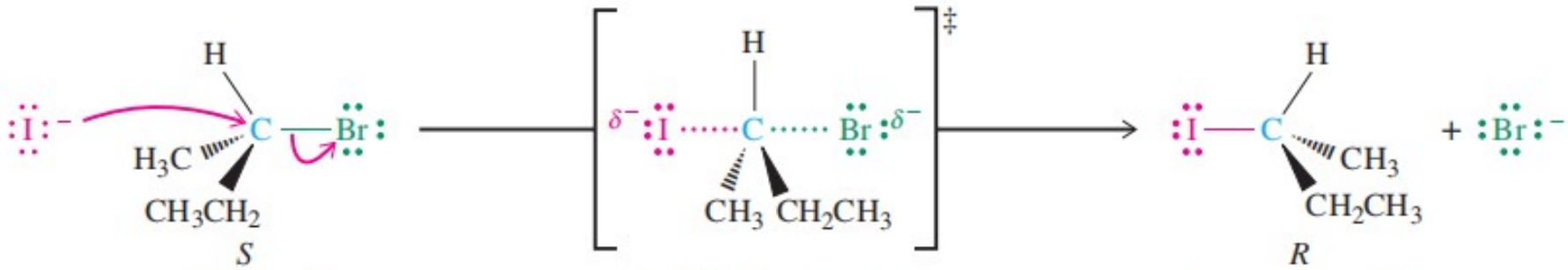
Neutral nucleophile gives positively charged product (as a salt)

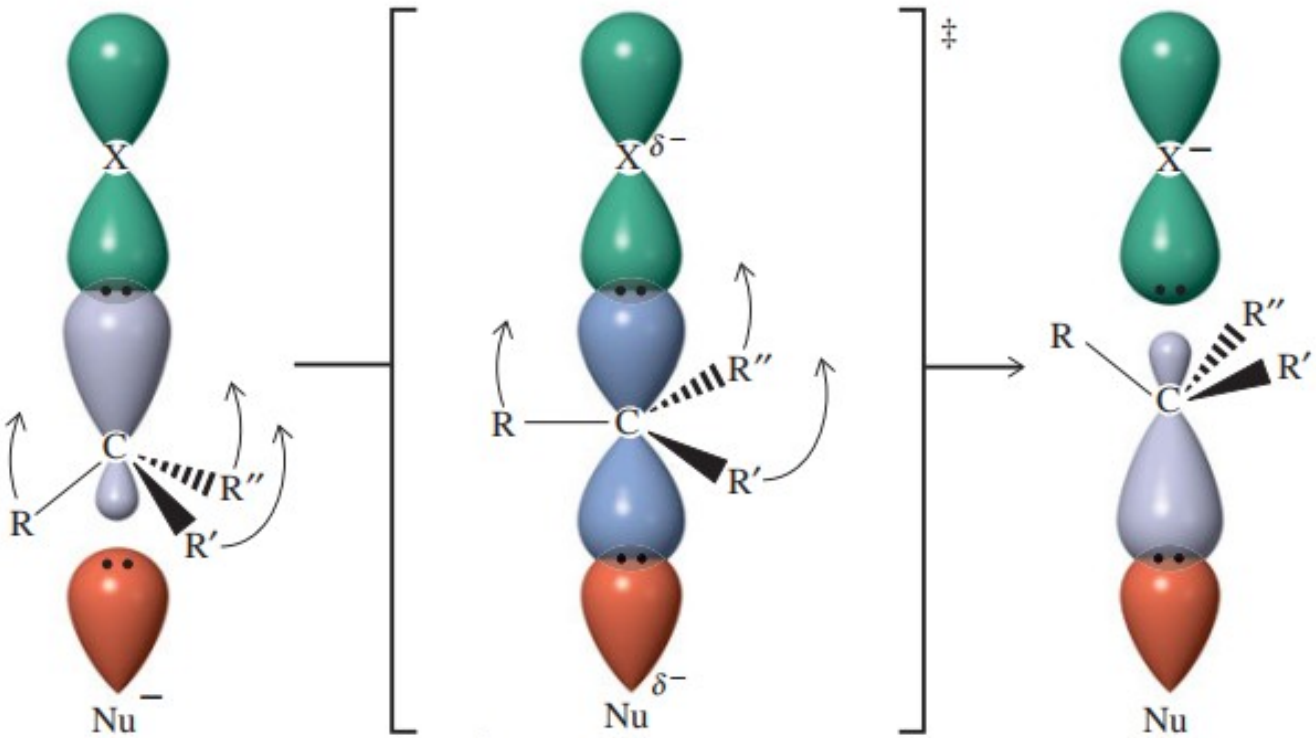


The SN2 reaction is stereospecifi c

- **Consider the reaction of (*S*)-2-bromobutane with iodide ion. Frontside displacement should give rise to 2-iodobutane with the *same* (*S*) confi guration as that of the substrate; backside displacement should furnish a product with the *opposite* confi guration.**
- **It is found that (*S*)-2-bromobutane gives (*R*)-2-iodobutane on treatment with iodide: *This and all other SN2 reactions proceed with* inversion of confi guration. A process whose mechanism requires that each stereoisomer of the starting material transform into a specifi c stereoisomer of product is described as stereospecific.**
- **The SN2 reaction is therefore stereospecific, proceeding by a backside displacement mechanism to give inversion of confi guration at the site of the reaction.**

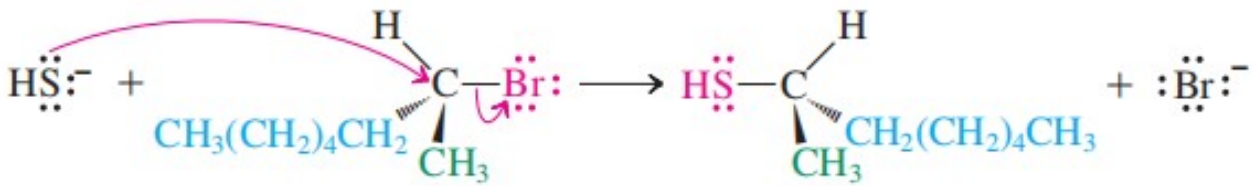
Stereochemistry of the Backside Displacement Mechanism for S_N2 Reactions





***sp*² hybridization at carbon**

Inversion of Configuration of an Optically Pure Compound by S_N2 Reaction

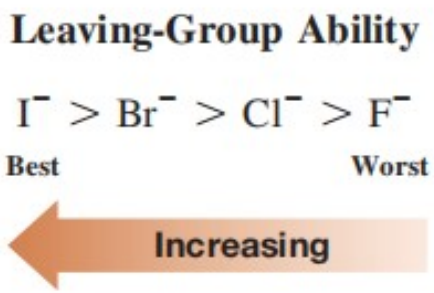


(*R*)-2-Bromooctane
([α] = −34.6)

(*S*)-2-Octanethiol
([α] = +36.4)

Leaving-group ability is a measure of the ease of its displacement

- nucleophilic substitution occurs only when the group being displaced, X, is readily able to depart, taking with it the electron pair of the C–X bond.
- the relative rate at which it can be displaced, its leaving-group ability, can be correlated with its capacity to accommodate a negative charge.
- Remember that a certain amount of negative charge is transferred to the leaving group in the transition state of the reaction .
- For the halogens, leaving-group ability increases along the series from fluorine to iodine.
- Thus, iodide is regarded as a "good" leaving group; fluoride, however, is so "poor" that SN2 reactions of fluoroalkanes are rarely observed.



Weak bases are good leaving groups

Leaving-group ability is inversely related to base strength. **Weak bases are best able to accommodate negative charge and are the best leaving groups. Among the halides, iodide is the weakest base and therefore the best leaving group in the series. Sulfates and sulfonates are weak bases as well.**

Table 6-4 Base Strengths and Leaving Groups

Conjugate acid		Leaving group	Conjugate acid		Leaving group
Strong	pK_a	*Good*	*Weak*	pK_a	*Poor*
HI (strongest)	−10.0	I^- (best)	HF	3.2	F^-
HBr	−9.0	Br^-	CH_3CO_2H	4.7	$CH_3CO_2^-$
HCl	−8.0	Cl^-	HCN	9.2	NC^-
H_2SO_4	−3.0	HSO_4^-	CH_3SH	10.0	CH_3S^-
H_3O^+	−1.7	H_2O	CH_3OH	15.5	CH_3O^-
CH_3SO_3H	−1.2	$CH_3SO_3^-$	H_2O	15.7	HO^-
			NH_3	35	H_2N^-
			H_2 (weakest)	38	H^- (worst)

Stronger acid ↑ (Strong → Weak)

Better leaving group ↑ (Good → Poor)

Stronger acid ↑ (Weak → Strong)

Better leaving group ↑ (Poor → Good)

Nucleophilicity decreases to the right in the periodic table

Experiments 1 and 2 compared pairs of nucleophiles containing the same nucleophilic element (e.g., oxygen in H_2O versus HO^- and nitrogen in H_3N versus H_2N^-). What about nucleophiles of similar structure but with different nucleophilic atoms? Let us examine the elements along one row of the periodic table.

Experiment 3

$$CH_3CH_2\ddot{\underset{..}{Br}}: + H_3N: \longrightarrow CH_3CH_2NH_3^+ + :\ddot{\underset{..}{Br}}:^- \quad \textbf{Fast}$$

$$CH_3CH_2\ddot{\underset{..}{Br}}: + H_2\ddot{\underset{..}{O}} \longrightarrow CH_3CH_2\ddot{O}H_2^+ + :\ddot{\underset{..}{Br}}:^- \quad \textbf{Very slow}$$

Experiment 4

$$CH_3CH_2\ddot{\underset{..}{Br}}: + H_2\ddot{N}:^- \longrightarrow CH_3CH_2\ddot{N}H_2 + :\ddot{\underset{..}{Br}}:^- \quad \textbf{Very fast}$$

$$CH_3CH_2\ddot{\underset{..}{Br}}: + H\ddot{\underset{..}{O}}:^- \longrightarrow CH_3CH_2\ddot{\underset{..}{O}}H + :\ddot{\underset{..}{Br}}:^- \quad \textbf{Slower}$$

Increasing basicity

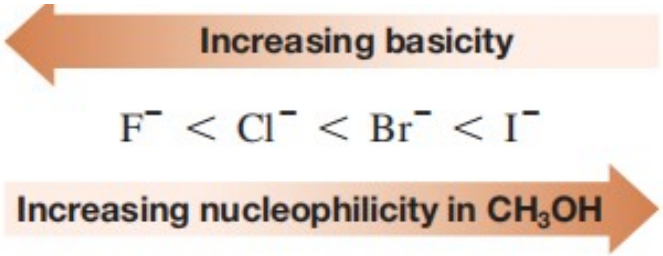
$$H_2N^- > HO^- > NH_3 > F^- > H_2O$$

Increasing nucleophilicity

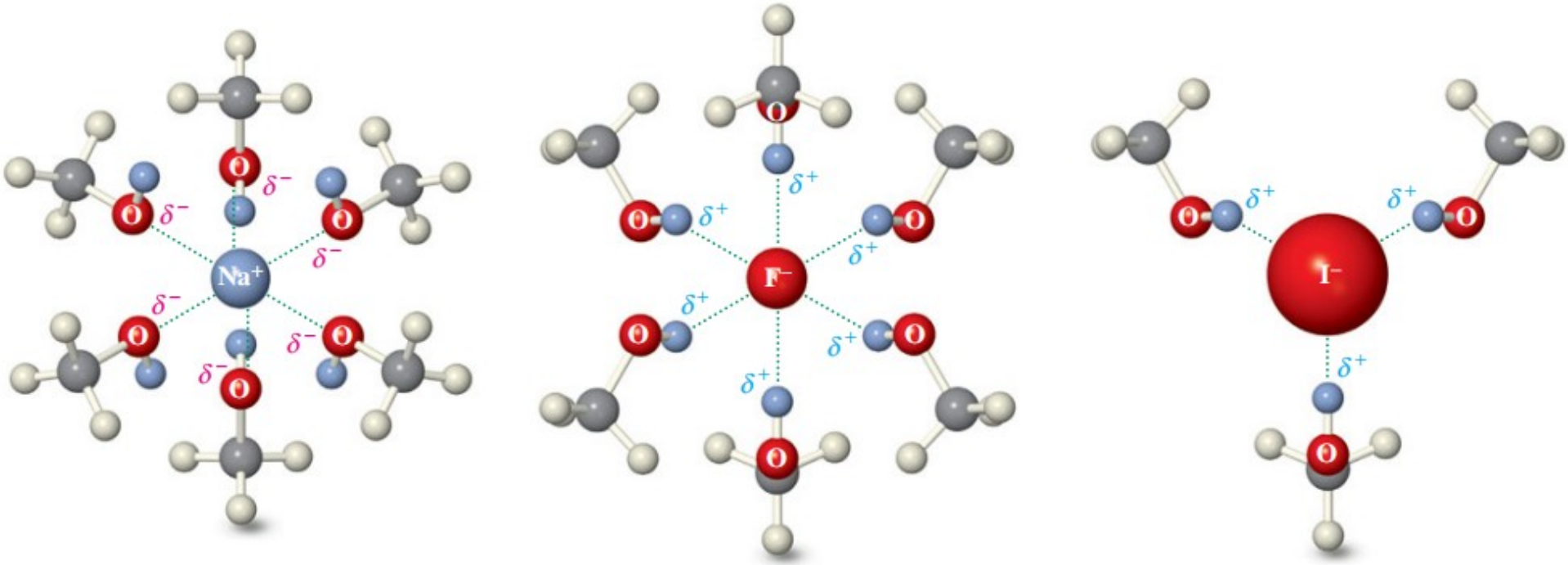
Solvation impedes nucleophilicity

- If it is a general rule that nucleophilicity correlates with basicity, then the elements considered from top to bottom of a column of the periodic table should show decreasing nucleophilic power that basicity decreases in an analogous fashion.
- To test this prediction, let us consider another series of experiments. In the equations below, we have explicitly added the solvent methanol to the reaction scheme, because, as we shall see, consideration of the solvent will be important in understanding the outcome of these experiments.

Surprisingly, nucleophilicity increases in the progression down the periodic table, a trend directly opposing that expected from the basicity of the nucleophiles tested. For example, in the series of halides, iodide is the fastest, although it is the weakest base.



Moving one column to the left in the periodic table, sulfi de nucleophiles are more reactive than the analogous oxide systems.


Na+
F−
I−
O
δ−
δ+

Aprotic solvents: the effect of solvation is diminished

- Other solvents that are useful in SN2 reactions are highly polar but aprotic.
- Several common examples are shown in Table 6-5; all lack protons capable of hydrogen bonding but do exhibit polarized bonds.
- Nitromethane even exists as a charge-separated species.
- Polar, aprotic solvents also dissolve salts by ion–dipole interactions, albeit not as well as protic solvents. Because they cannot form hydrogen bonds, they solvate anionic nucleophiles relatively weakly. The consequences are twofold. First, compared to protic solvents, the *reactivity of the nucleophile is raised,* sometimes dramatically.
- For example, bromomethane reacts withpotassium iodide 500 times faster in acetone than in methanol.
- Table 6-6 compares the rates of SN2 reactions of iodomethane with chloride in three protic solvents—methanol, formamide, and N-methylformamide—and one aprotic solvent, N,N-dimethylformamide (DMF). (Formamide and N-methylformamide can form hydrogen bonds by virtue of their polarized N–H linkages.)
- The rate of reaction in DMF is more than a million times greater than it is in methanol.

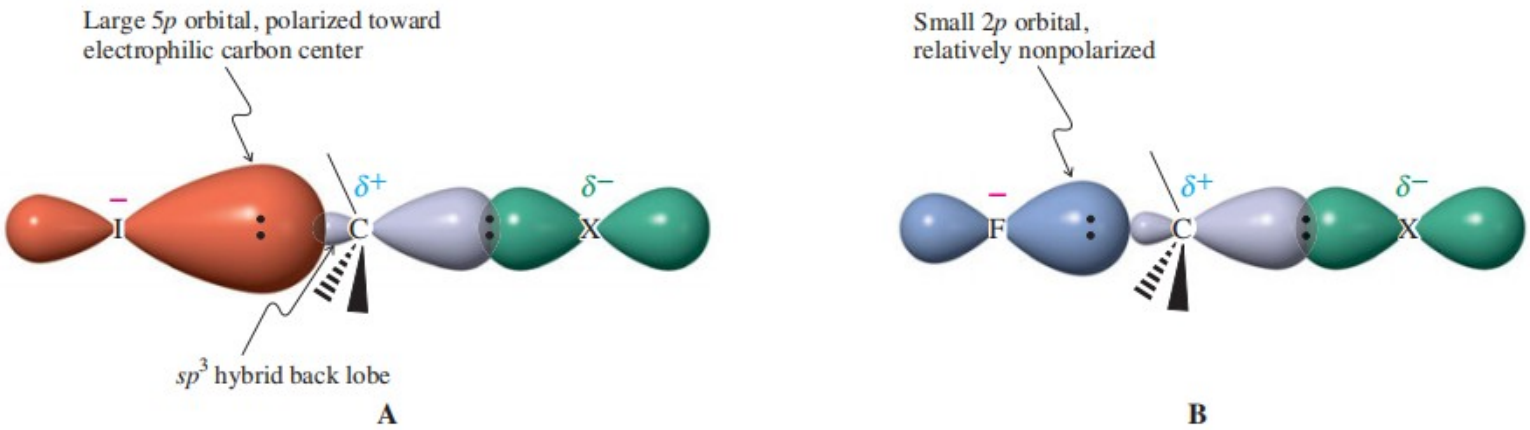
Table 6-6 Relative Rates of S_N2 Reactions of Iodomethane with Chloride Ion in Various Solvents

$$CH_3I + Cl^- \xrightarrow[k_{rel}]{\text{Solvent}} CH_3Cl + I^-$$

Solvent			
Formula	Name	Classification	Relative rate (k_{rel})
CH_3OH	Methanol	Protic	1
$HCONH_2$	Formamide	Protic	12.5
$HCONHCH_3$	*N*-Methylformamide	Protic	45.3
$HCON(CH_3)_2$	*N,N*-Dimethylformamide	Aprotic	1,200,000

Increasing polarizability improves nucleophilic power

- **The solvation effects just described should be very pronounced only for charged nucleophiles.**
- **Nevertheless, the degree of nucleophilicity increases down the periodic table, even for *uncharged nucleophiles,* for which solvent effects should be much less strong, for example, H_2Se >H_2S >H_2O, and PH_3 >NH_3.**
- **Therefore, there must be an additional factor that comes into play.**
- **This factor is the polarizability of the nucleophile (Section 6-1).**
- **Larger elements have larger, more diffuse, and more polarizable electron clouds. These electron clouds allow for more effective orbital overlap in the SN2 transition state (Figure). The result is a lower transition-state energy and faster nucleophilic substitution.**



Branching at the reacting carbon decreases the rate of the SN2 reaction

experiments show that reactivities decrease rapidly in the order shown in Table 6-8. We can find an explanation by comparing the transition states for these four substitutions. Figure 6-8A shows this structure for the reaction of chloromethane with hydroxide ion. The carbon is surrounded by the incoming nucleophile, the outgoing leaving group, and three substituents (all hydrogen in this case). Although the presence of these five groups increases the crowding about the carbon relative to that in the starting halomethane, the hydrogens do not give rise to serious steric interactions with the nucleophile because of their small size. However, replacement of one hydrogen by a methyl group, as in a halo ethane, creates substantial steric repulsion with the incoming nucleophile, thereby raising the transition-state energy (Figure 6-8B). This effect signifi cantly retards nucleophilic attack.

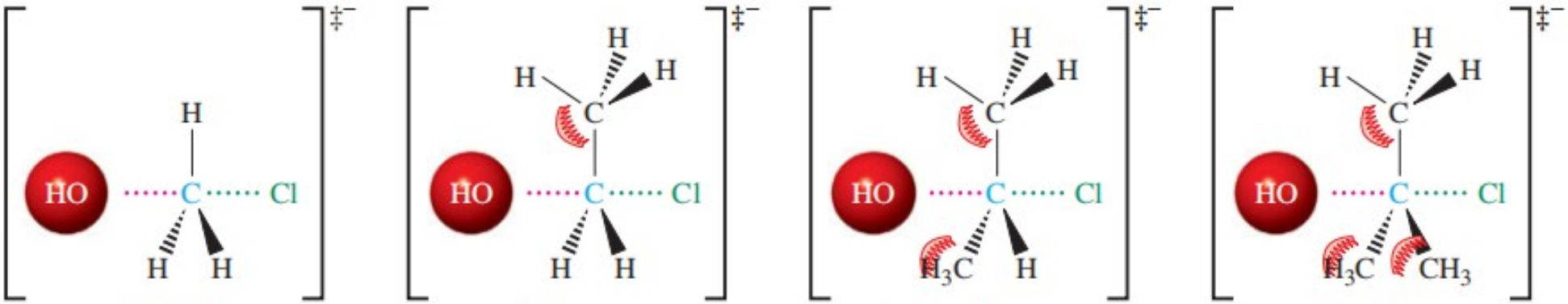
Table 6-8

Relative Rates of S_N2 Reaction of Branched Bromoalkanes with Iodide

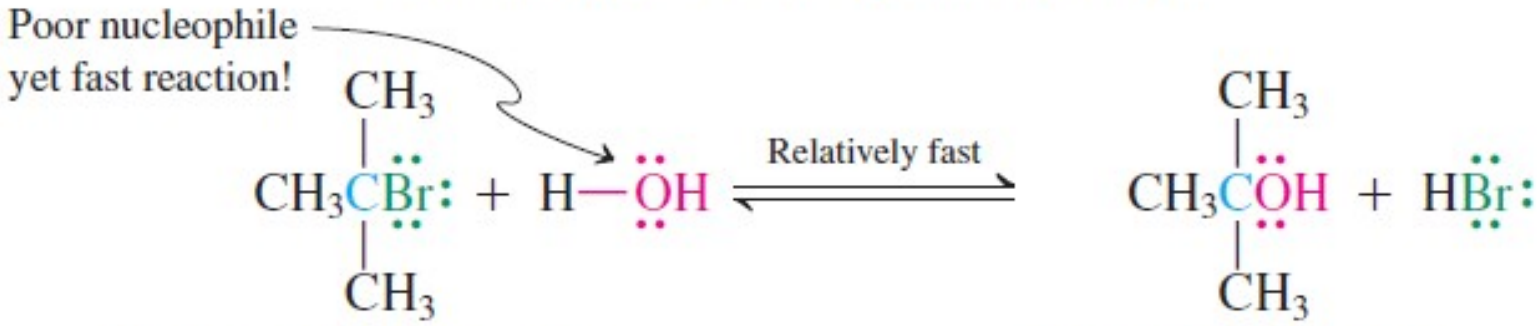
Bromoalkane	Rate
CH_3Br	145
CH_3CH_2Br	1
$CH_3CH(CH_3)Br$	0.0078
$CH_3C(CH_3)_2Br$	Negligible

Increasing size of R

Decreasing reactivity in S_N2 reaction


HO
C
Cl
H
H
H

An Example of Solvolysis: Hydrolysis

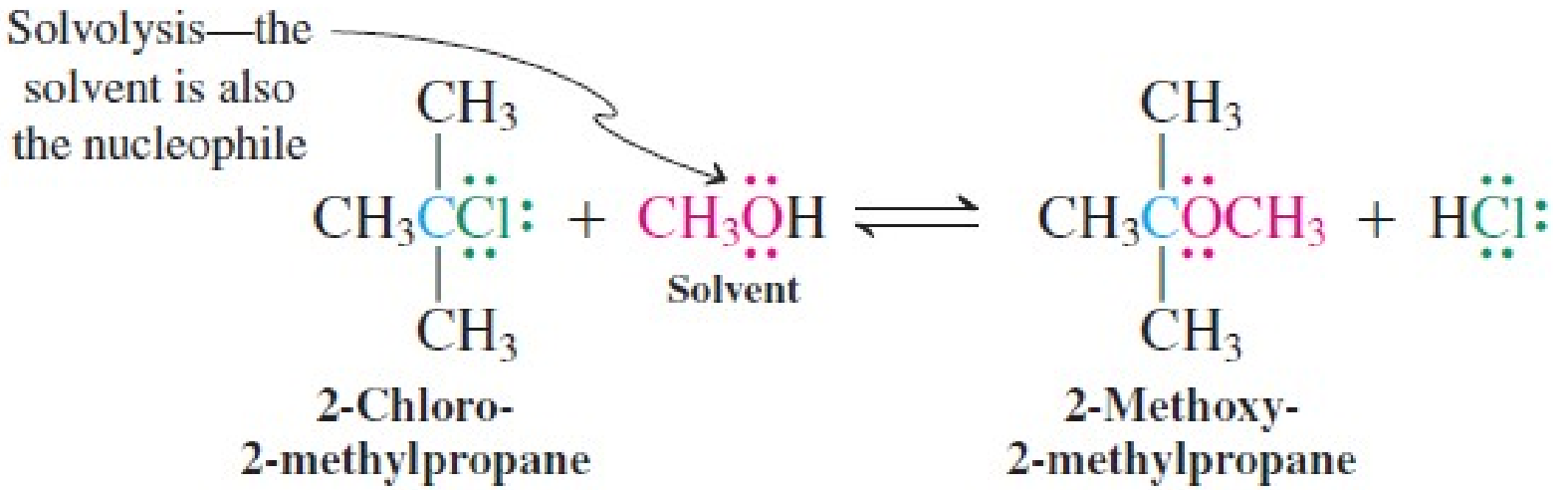


2-Bromo-2-methylpropane
(*tert*-Butyl bromide)

2-Methyl-2-propanol
(*tert*-Butyl alcohol)

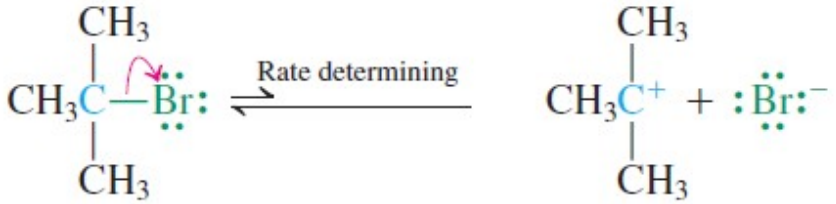
Solvolysis also takes place in alcohol solvents.

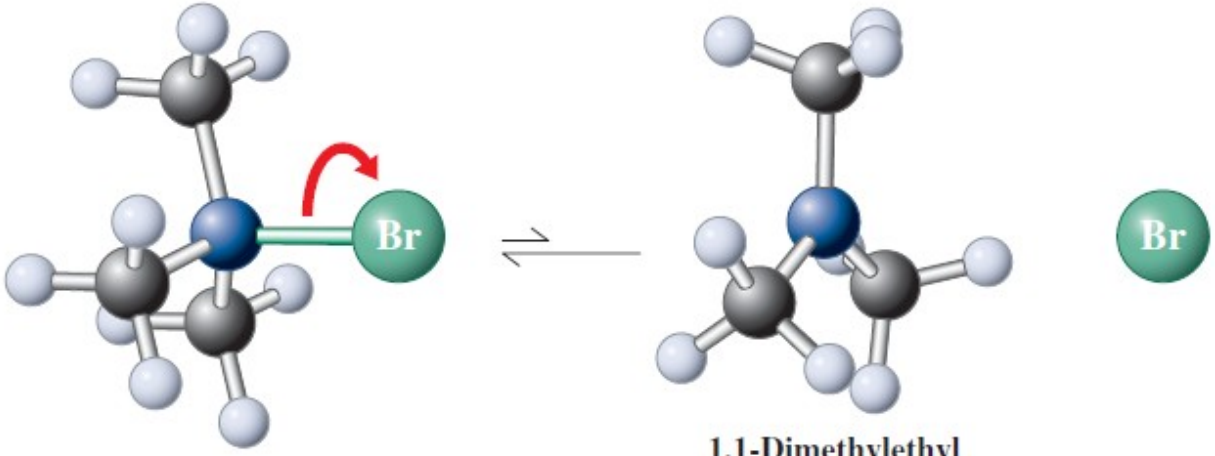
Solvolysis of 2-Chloro-2-methylpropane in Methanol



Step 1. The rate-determining step is the dissociation of the haloalkane to an alkyl cation and bromide.This conversion is an example of heterolytic cleavage. The hydrocarbon product contains a positively charged central carbon atom attached to three other groups and bearing onlyan electron sextet. Such a structure is called a carbocation

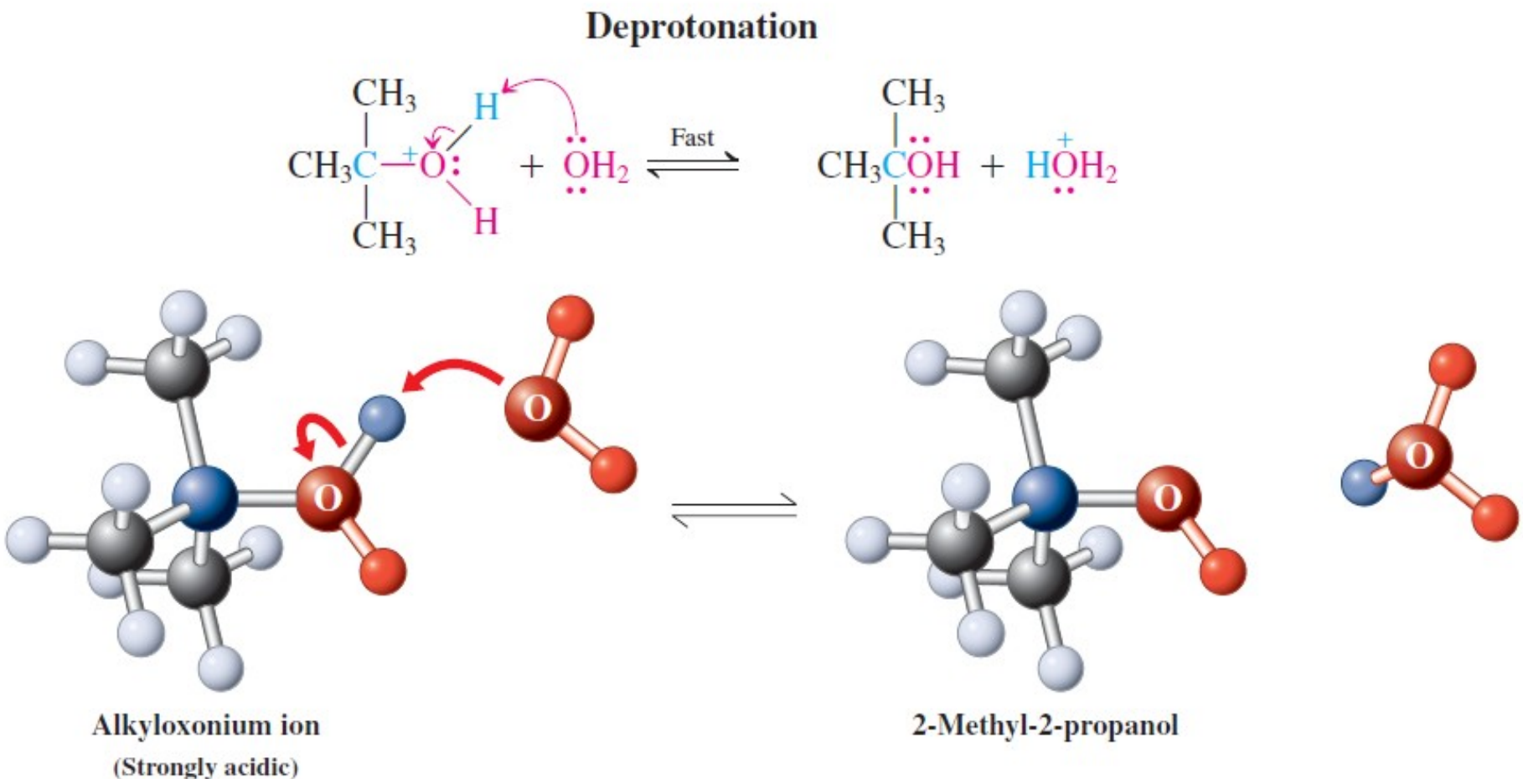
Dissociation of Halide to Form a Carbocation



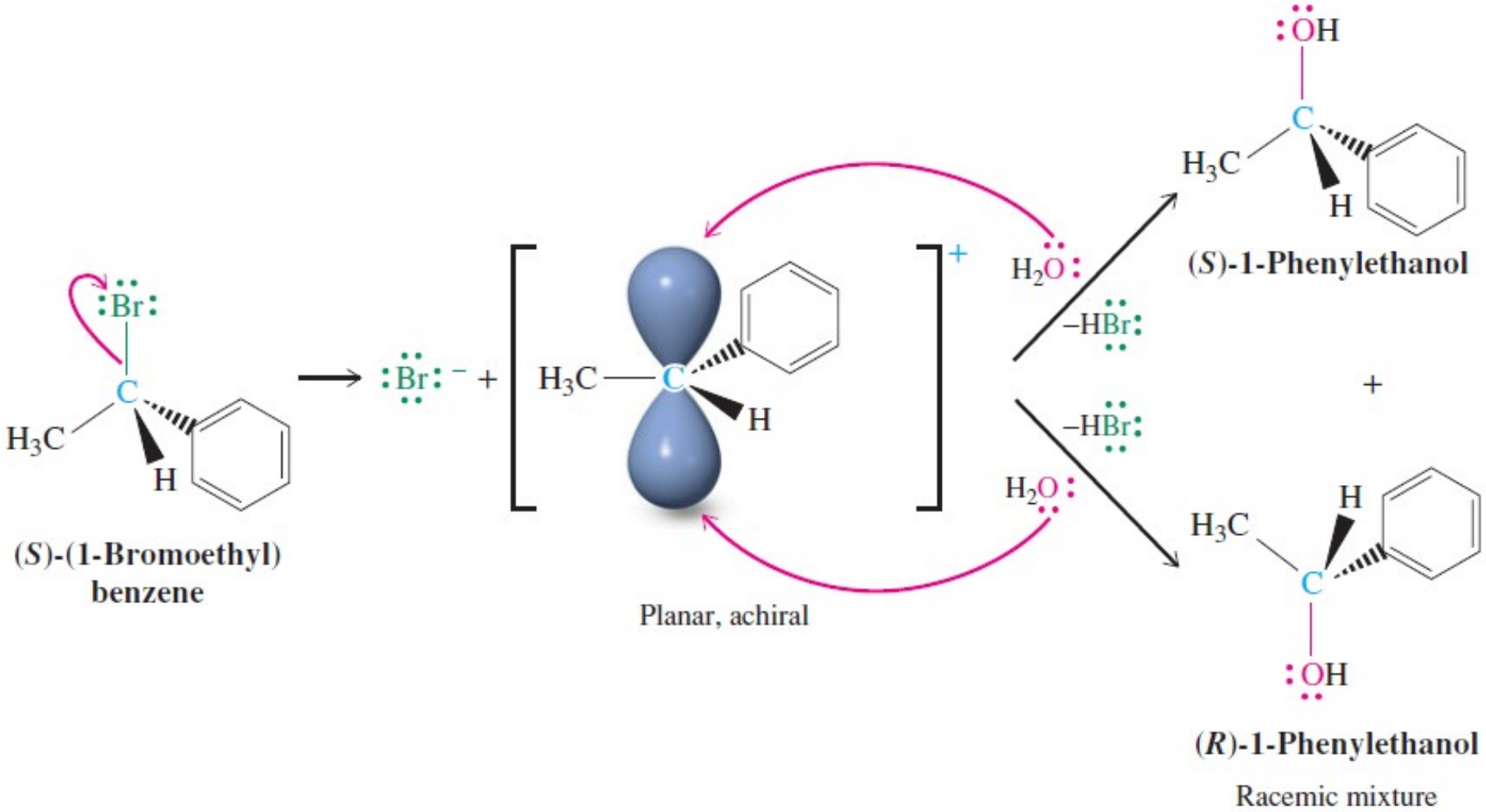


1,1-Dimethylethyl cation (*tert*-Butyl cation)

Step 3. Like the hydronium ion, H_3O, the first member of the series of oxonium ions,all alkyloxonium ions are strong acids. They are therefore readily deprotonated by the water in the reaction medium to furnish the fi nal alcohol.



All three steps of the mechanism of solvolysis are reversible. The overall equilibrium can be driven in either direction by the suitable choice of reaction conditions. Thus, a large excess of nucleophilic solvent ensures complete solvolysis.


(S)-(1-Bromoethyl) benzene
Planar, achiral
(S)-1-Phenylethanol
(R)-1-Phenylethanol
Racemic mixture

Effect of Solvent on the Rate of an S_N1 Reaction

			Relative rate
$(CH_3)_3CBr$	$\xrightarrow[\text{More polar solvent}]{100\%\ H_2O}$	$(CH_3)_3COH + HBr$	400,000
$(CH_3)_3CBr$	$\xrightarrow[\text{Less polar solvent}]{90\%\ \text{acetone},\ 10\%\ H_2O}$	$(CH_3)_3COH + HBr$	1

The SN1 reaction speeds up with better leaving groups

- **Because the leaving group departs in the rate-determining step of the SN1 reaction, it is not surprising that the rate of the reaction increases as the leaving-group ability of the departing group improves.**
- **Thus, tertiary iodoalkanes more readily undergo solvolysis than do the corresponding bromides, and bromides are, in turn, more reactive than chlorides. Sulfonates are particularly prone to departure.**

Relative Rate of Solvolysis of RX (R = Tertiary Alkyl)

$$X = -OSO_2R' > -I > -Br > -Cl$$

Increasing rate

- **For example, hydrolysis of a solution of 2-chloro-2-methylpropane gives the expected 2-methyl-2-propanol (tert-butyl alcohol), with a rate constant k1.**
- **Quite a different result is obtained when the same experiment is carried out in the presence of the soluble salt calcium formate: 1,1-Dimethylethyl formate (tert-butyl formate) replaces the alcohol as the product, but the reaction still proceeds at exactly the same rate k1.**
- **In this case, formate ion, a better nucleophile than water, wins out in competition for bonding to the intermediate carbocation. The rate of disappearance of the starting material is determined by k1**

(regardless of the product eventually formed), but the relative yields of the products depend on the relative reactivities and concentrations of the competing nucleophiles.

Effect of the Alkyl Group on the SN1 Reaction: Carbocation Stability
What is so special about tertiary haloalkanes

❖ **Carbocation stability increases from primary to secondary to tertiary.**

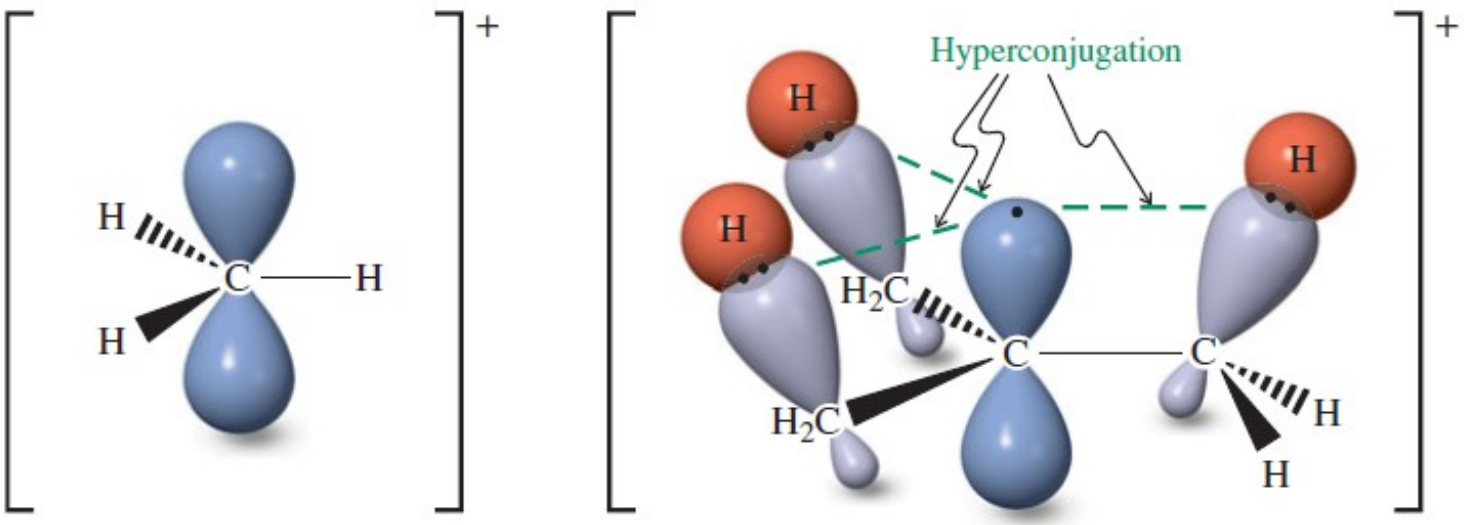
Relative Stability of Carbocations

$$CH_3CH_2CH_2\overset{+}{C}H_2 < CH_3CH_2\overset{+}{C}HCH_3 < (CH_3)_3\overset{+}{C}$$

Primary < Secondary < Tertiary

Increasing carbocation stability

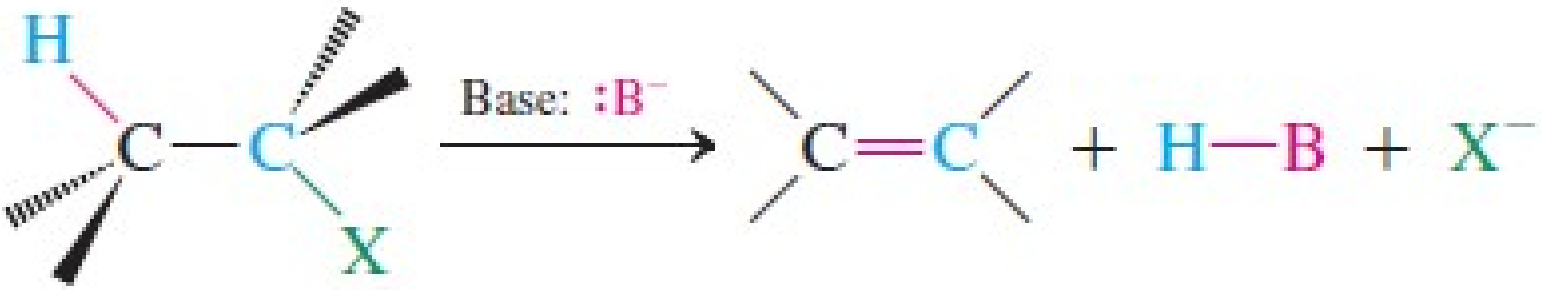
❖ **Hyperconjugation stabilizes positive charge.**



A Methyl cation

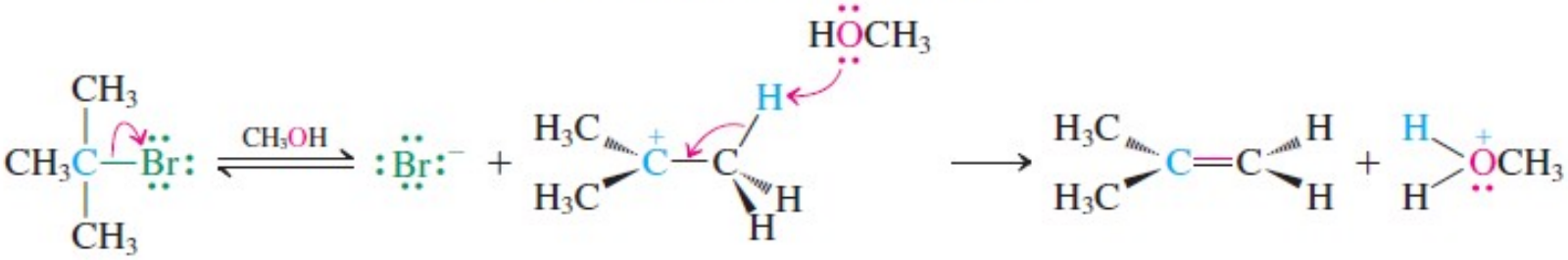
B 1, 1-Dimethylethyl cation (*tert*-Butyl cation)

Elimination

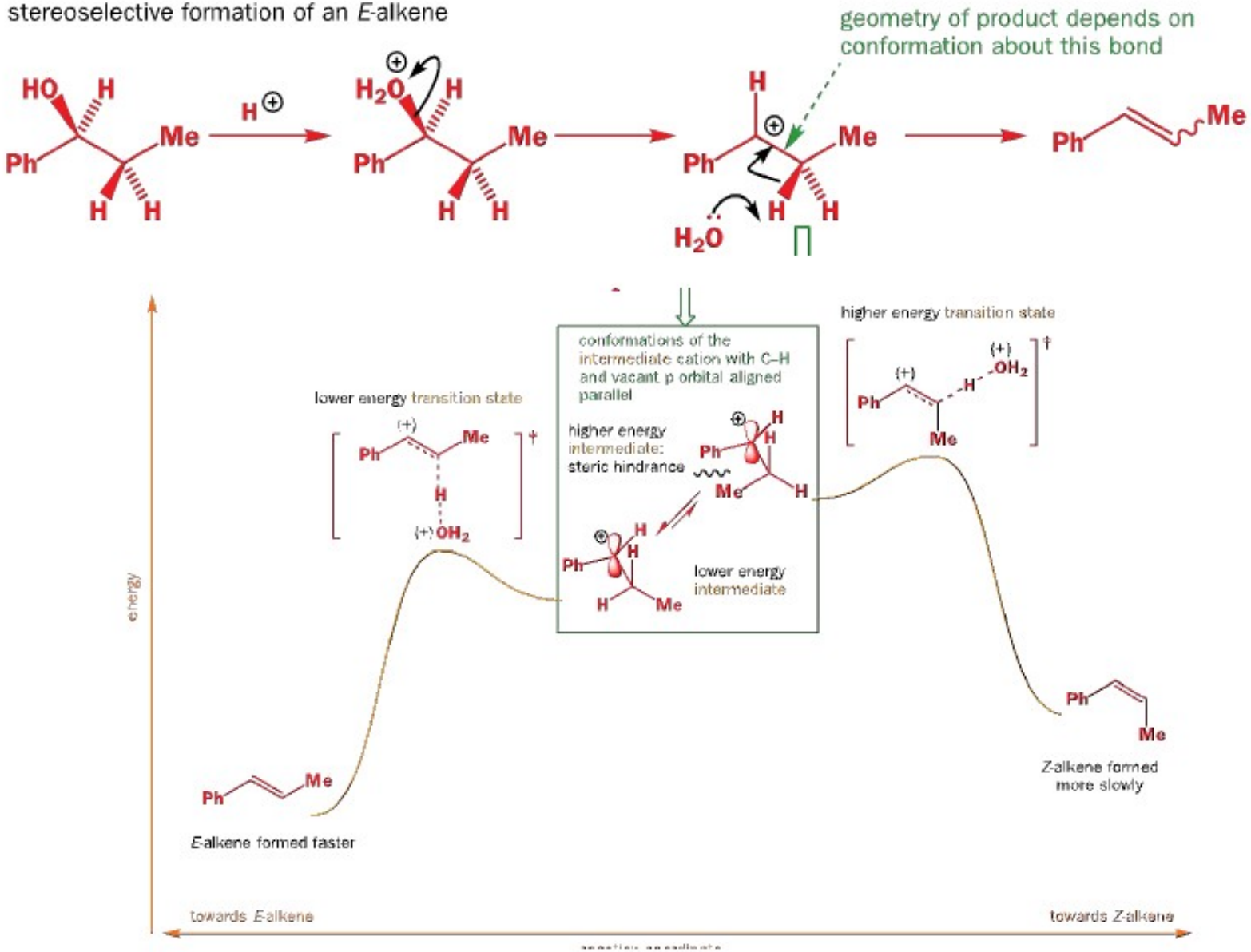


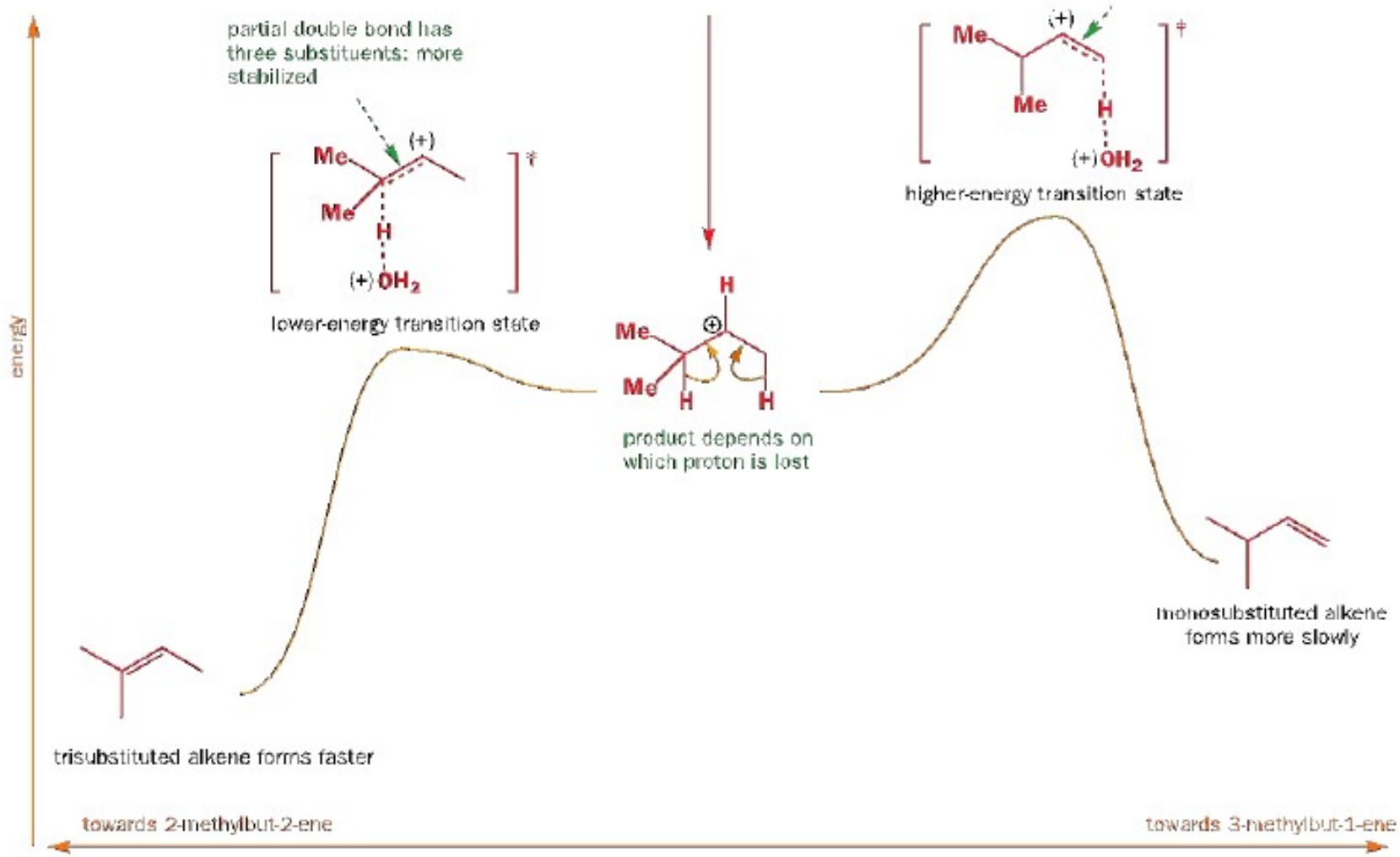
The rate-determining step in the E1 process is the same as that in SN1 reactions: dissociation to a carbocation. This intermediate then has a second pathway at its disposal along with nucleophilic rapping: loss of a proton from a carbon adjacent to the one bearing the positive charge.

The E1 Reaction Mechanism



stereoselective formation of an *E*-alkene




partial double bond has three substituents: more stabilized
Me
Me
(+)
H
(+) OH2
lower-energy transition state
Me
(+)
Me
H
(+) OH2
higher-energy transition state
H
Me
Me
H
H
product depends on which proton is lost
energy
monosubstituted alkene forms more slowly
trisubstituted alkene forms faster
towards 2-methylbut-2-ene
towards 3-methylbut-1-ene

- **The nature of the leaving group should have no effect on the ratio of substitution to elimination, because the carbocation formed is the same in either case. This is indeed observed (Table 7-3).**
- **The product ratio may be affected by the addition of base, but at low base concentration this effect is usually small.**
- **Recall that strong bases are usually strong nucleophiles as well , so addition of a base will generally not greatly favor deprotonation of the carbocation at the expense of nucleophilic attack, and the ratio of E1 to SN1 products remains approximately constant.**
- **Indeed, elimination by the E1 mechanism is usually no more than a minor side reaction accompanying SN1 substitution.**
- **Is there a way to make elimination the major outcome, to give alkenes as major products?**
- **Yes: By using high concentrations of strong base, the proportion of elimination rises dramatically. This effect does not arise from a change in the E1:SN1 ratio.**

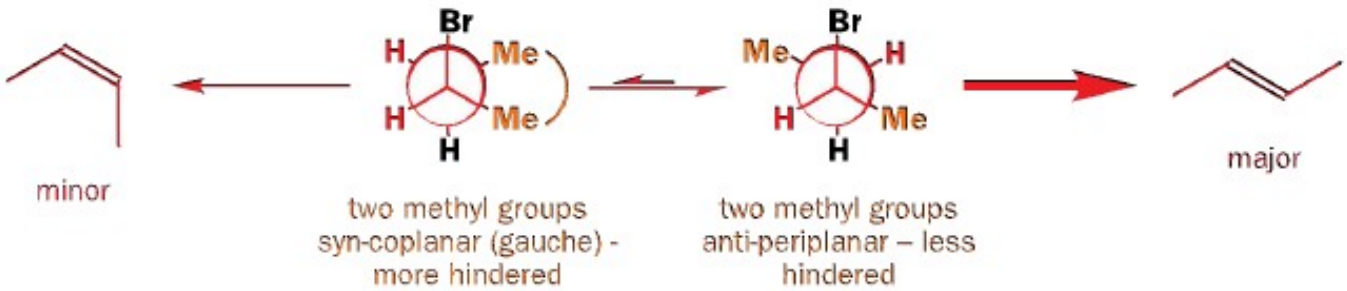
Kinetics of the E2 Reaction of 2-Chloro-2-methylpropane

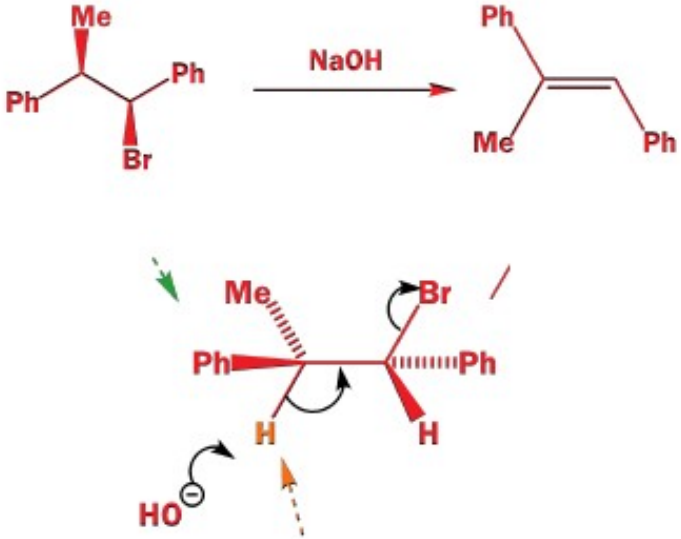
$$(CH_3)_3CCl + Na^+\,^-OH \xrightarrow{k} CH_2{=}C(CH_3)_2 + NaCl + H_2O$$

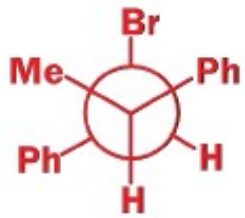
$$\text{Rate} = k[(CH_3)_3CCl][^-OH]\ \text{mol L}^{-1}\ \text{s}^{-1}$$

The preceding section taught us that unimolecular elimination may compete with substitution.However, a dramatic change of the kinetics is observed at higher concentrations of strong base. The rate of alkene formation becomes proportional to the concentrations of both the starting halide and the base: The kinetics of elimination are now second order and the process is called bimolecular elimination, abbreviated E2.

H and Br must be anti-periplanar for E2 elimination:
two possible conformations

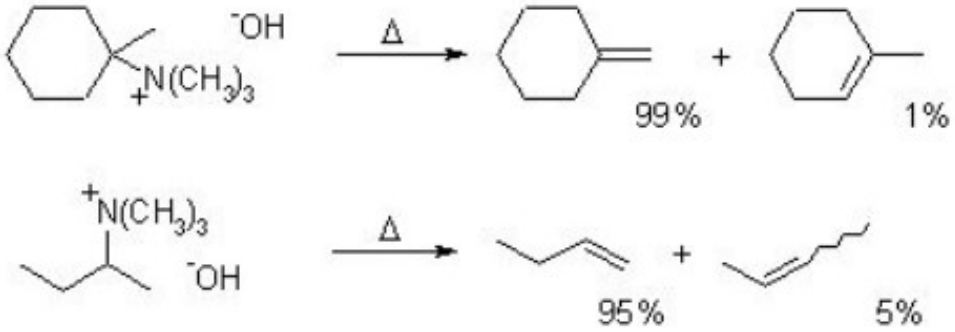




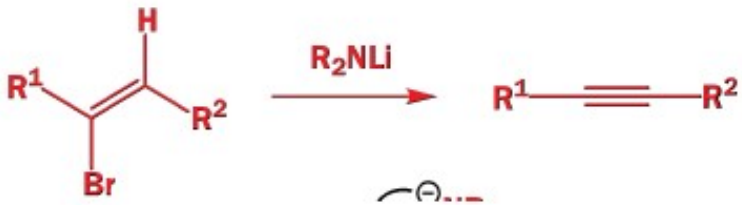


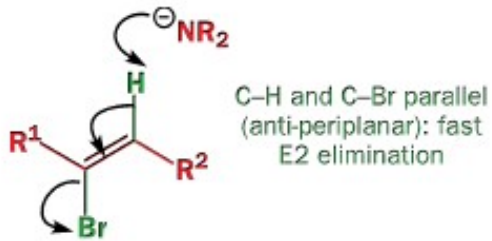
H and Br must be
anti-periplanar

Hofmann's Rule: The major elimination product is the less highly substituted alkene

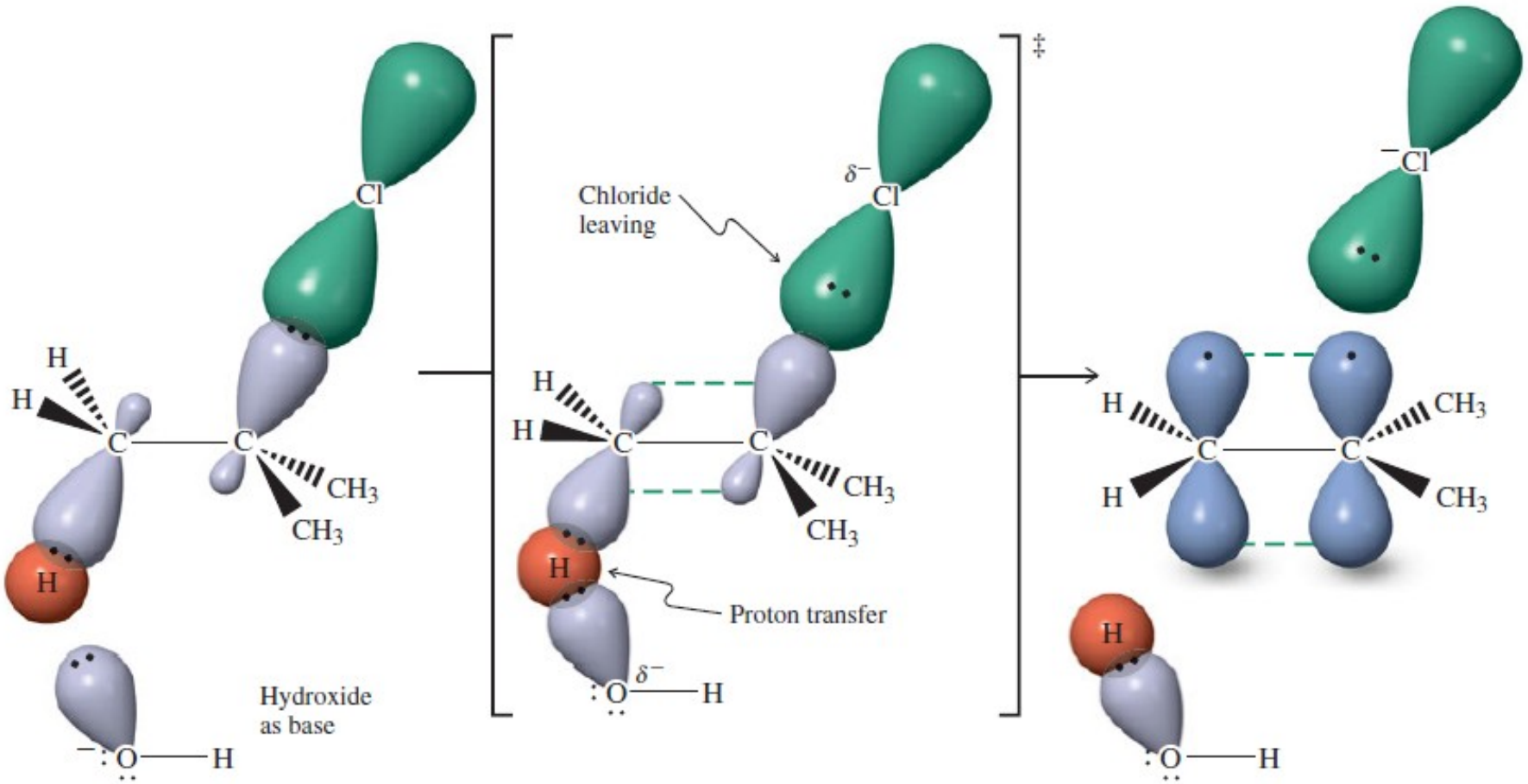


Preparation of Alkyne: E2 elimination from vinyl halides

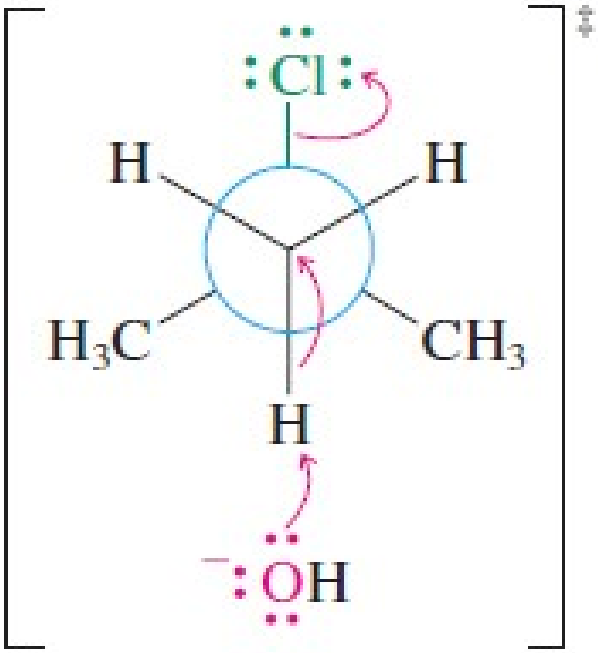




1. Deprotonation by the base
2. Departure of the leaving group
3. Rehybridization of the reacting carbon centers from sp3 to sp2 to furnish the two p orbitals of the emerging double bond.All three changes take place simultaneously: The E2 is a one-step, concerted process.



Newman Projection of E2 Transition State

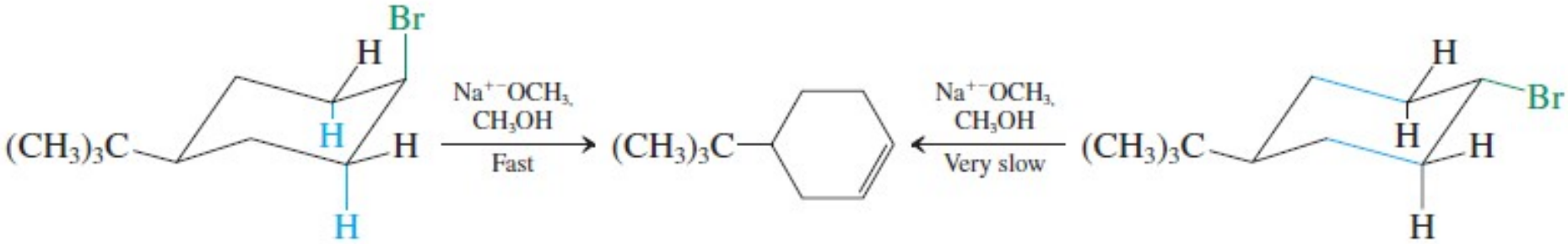


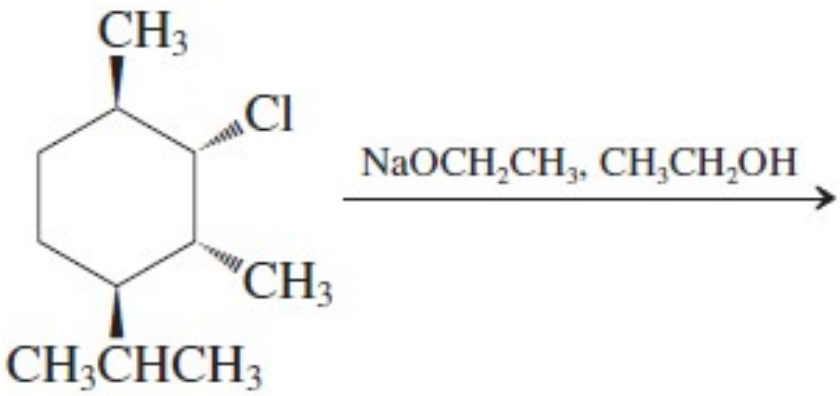
Relative Reactivity in the E2 Reaction

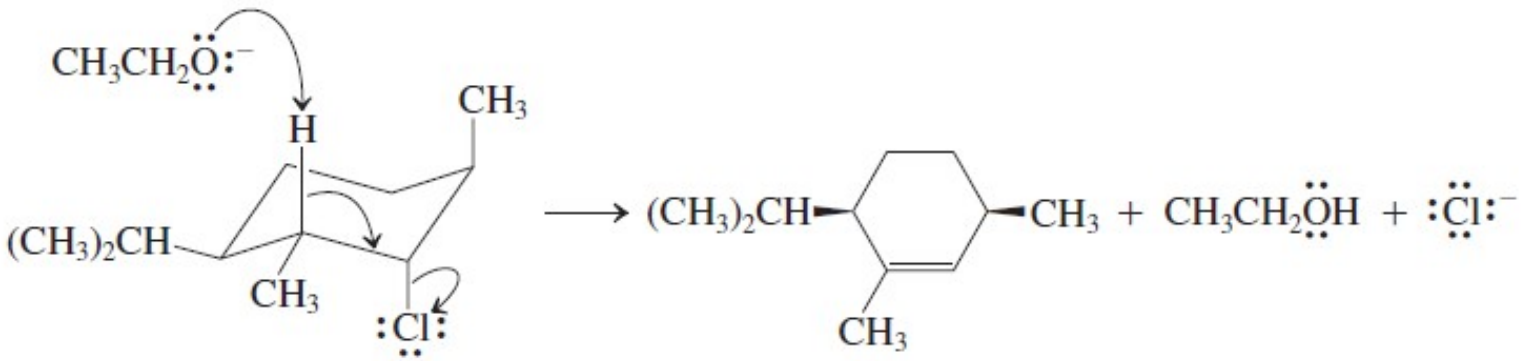
RI > RBr > RCl > RF

Increasing

Anti Elimination Occurs Readily for _cis_- but Not for _trans_-1-Bromo-4-(1,1-dimethylethyl)cyclohexane

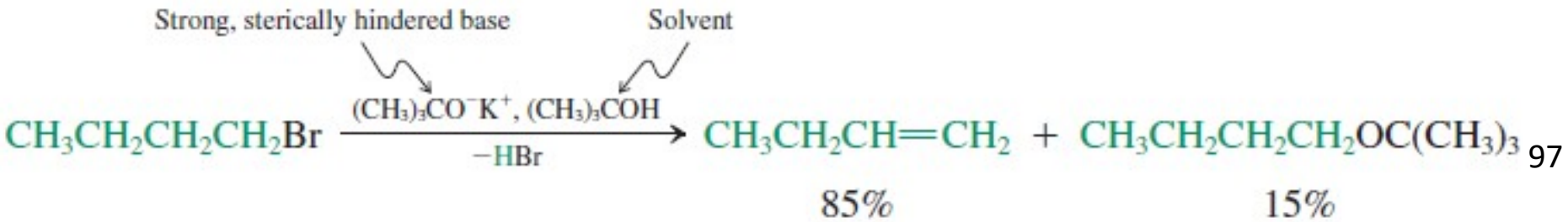



CH3
Cl
CH3
CH3CHCH3
NaOCH2CH3, CH3CH2OH


CH3CH2O:−
H
CH3
(CH3)2CH
CH3
:Cl:
(CH3)2CH
CH3
CH3
+ CH3CH2OH + :Cl:−

Sterically hindered basic nucleophiles favor elimination

- primary haloalkanes react by substitution with good nucleophiles, including strong bases.
- The situation changes when the steric bulk of the nucleophile hinders attack at the electrophilic carbon.
- In this case, elimination may predominate, even with primary systems, through deprotonation at the less hindered periphery of the molecule.
- Two examples of sterically hindered bases that are frequently employed in eliminations are potassium tert-butoxide and lithium diisopropylamide (LDA).
- The first contains a tertiary alkyl group on oxygen; the second has two secondary alkyl groups on nitrogen.
- For use in such reactions, these bases are frequently dissolved in their conjugate acids, 2-methyl-2-propanol (tert-butyl alcohol) and 1-methyl-N-(1-methylethyl)ethanamine (diisopropylamine), respectively.



90% yield of 2:1 mixture of *E*;*Z* alkenes

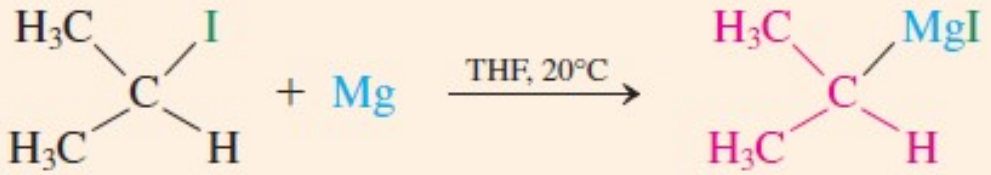
- **Alkyllithium compounds and Grignard reagents are rarely isolated; they are formed in solution and used immediately in the desired reaction.**
- **Sensitive to air and moisture, they must be prepared and handled under rigorously air- and water-free conditions. Simple examples, such as methyllithium, methylmagnesium bromide, butyllithium, and others, are commercially available.**
- **The formulas RLi and RMgX oversimplify the true structures of these reagents. Thus, as written, the metal ions are highly electron deficient.**
- **To make up the desired electron octet, they function as Lewis acids and attach themselves to the Lewis basic solvent molecules.**
- **For example, alkylmagnesium halides are stabilized by bonding to two ether molecules. The solvent is said to be coordinated to the metal. This coordination is rarely shown in equations, but it is crucial for the formation of the Grignard species.**

Alkyllithium Synthesis

$$CH_3Br + 2\ Li \xrightarrow{(CH_3CH_2)_2O,\ 0^\circ - 10^\circ C} CH_3Li + LiBr$$

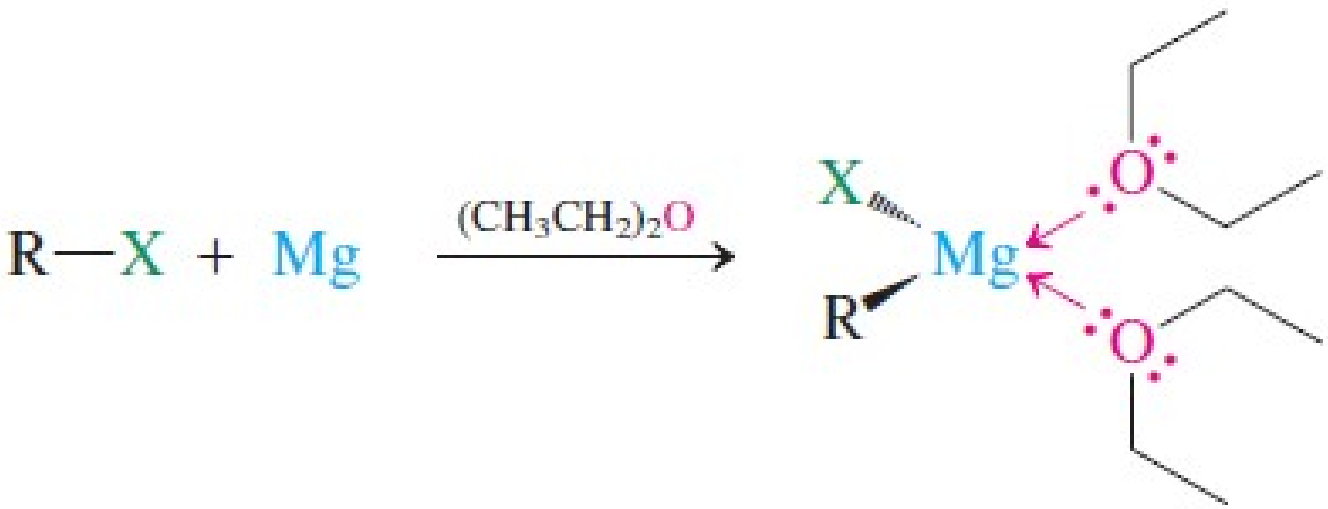
Methyl-
lithium

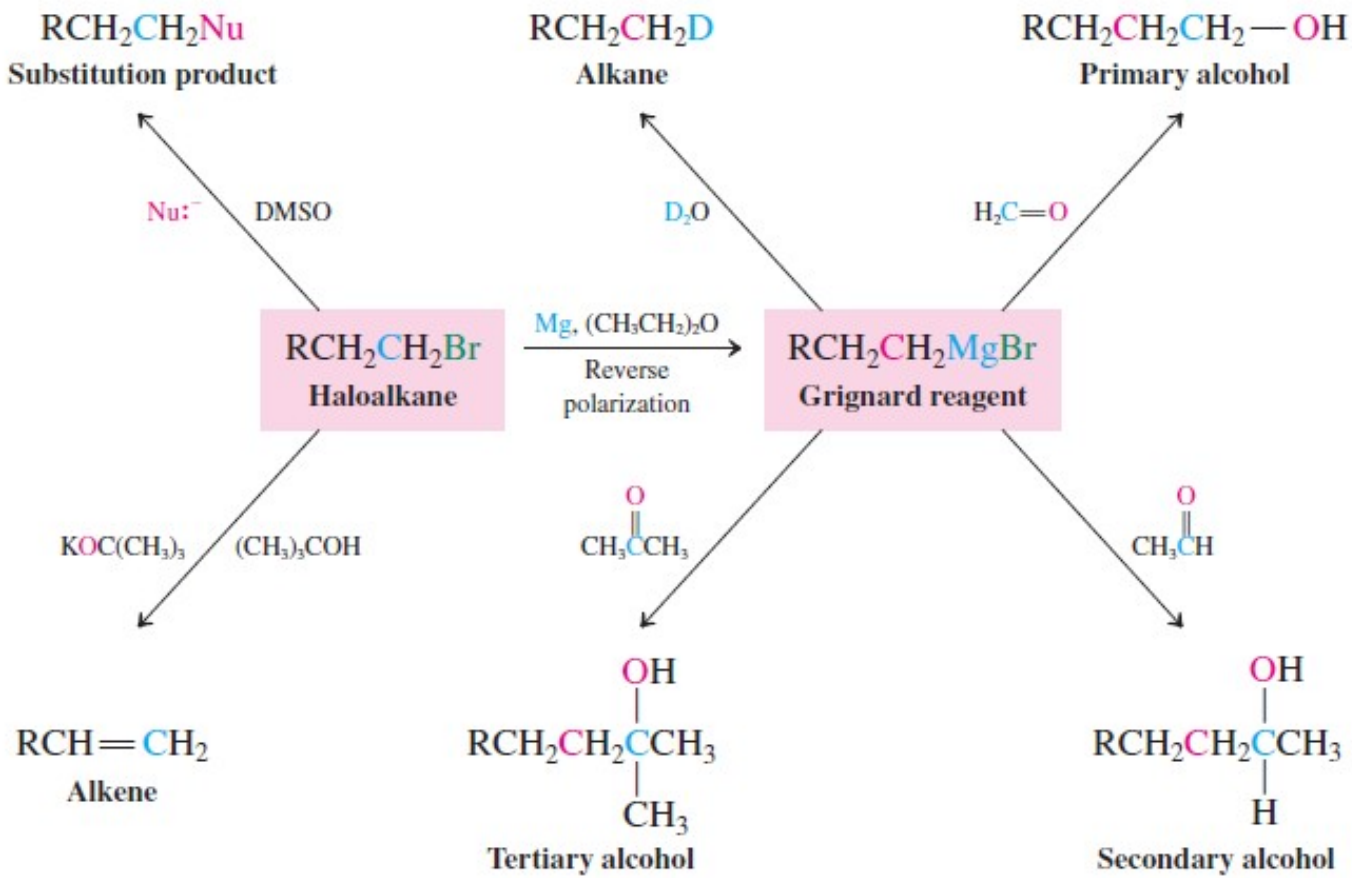
Alkylmagnesium (Grignard) Synthesis



**1-Methylethyl-
magnesium iodide**

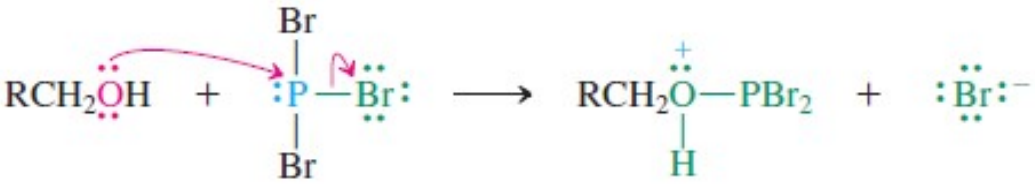
Grignard Reagents Are Coordinated to Solvent




RCH2CH2Nu
Substitution product
RCH2CH2D
Alkane
RCH2CH2CH2—OH
Primary alcohol
Nu:⁻
DMSO
D2O
H2C=O
RCH2CH2Br
Haloalkane
Mg, (CH3CH2)2O
Reverse polarization
RCH2CH2MgBr
Grignard reagent
KOC(CH3)3
(CH3)3COH
CH3CCH3
CH3CH
RCH=CH2
Alkene
RCH2CH2CCH3
Tertiary alcohol
Secondary alcohol

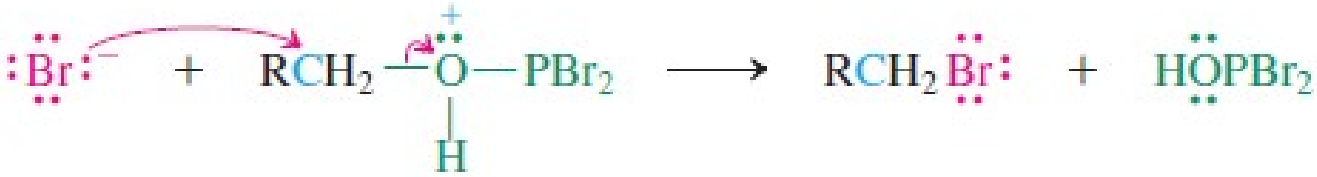
In the fi rst step, the alcohol attacks the phosphorus reagent as a nucleophile to form a protonated inorganic ester, a derivative of phosphorous acid.

Step 1



Next, $HOPBr_2$, a good leaving group, is displaced (SN_2) by the bromide generated in step 1, producing the haloalkane.This method of haloalkane synthesis is especially efficient because $HOPBr_2$ continues to react successively with two more molecules of alcohol, converting them into haloalkane as well.

Step 2

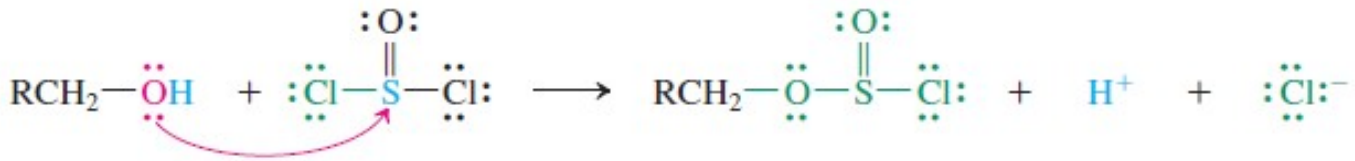


A chlorinating agent commonly used to convert alcohols into chloroalkanes is thionyl chloride, SOCl2. Simply warming an alcohol in its presence results in the evolution of SO2 and HCl and the formation of the chloroalkane

Chloroalkane Synthesis with $SOCl_2$

$$CH_3CH_2CH_2OH + SOCl_2 \longrightarrow \underset{91\%}{CH_3CH_2CH_2Cl} + O{=}S{=}O + HCl$$

Step 1

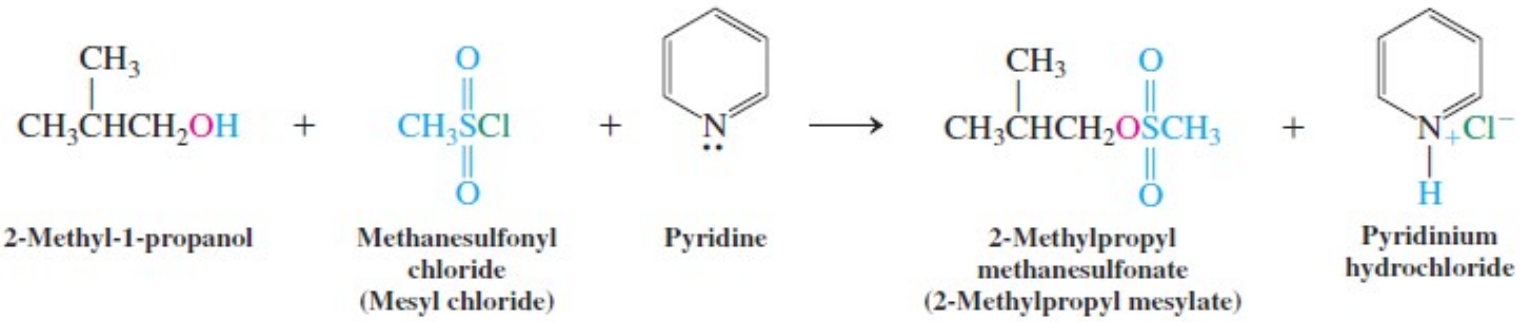


The chloride ion created in this process then acts as a nucleophile and attacks the ester, an SN2 reaction yielding one molecule each of SO2 and HCl.

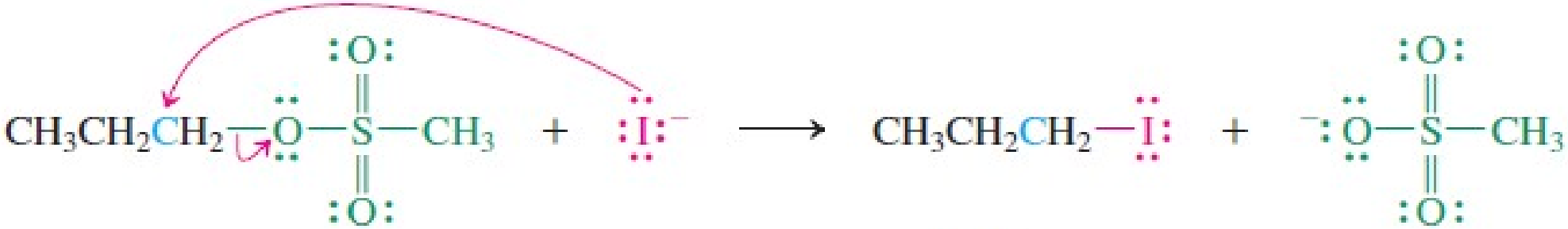
Alkyl sulfonates are versatile substrates for substitution reactions

- The inorganic esters in the reactions of SOCl2 are special examples of leaving groups derived from sulfur-based acids.
- They are related to the sulfonates . Alkyl sulfonates contain excellent leaving groups and can be readily prepared from the corresponding sulfonyl chlorides and an alcohol.
- A mild base such as pyridine or a tertiary amine is often added to remove the HCl formed.

Synthesis of an Alkyl Sulfonate



Substitution Reactions of Alkyl Sulfonates

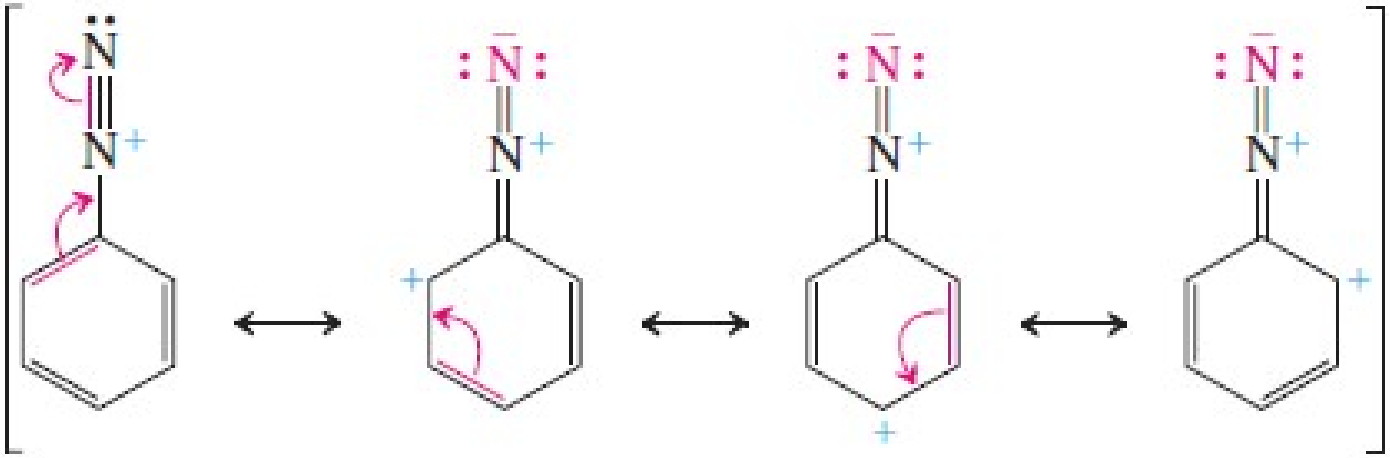


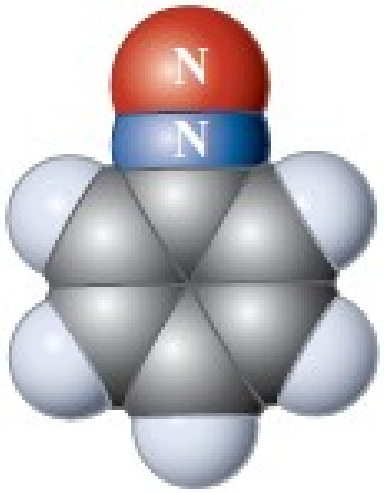
90%

Arenediazonium salts are stabilized by resonance

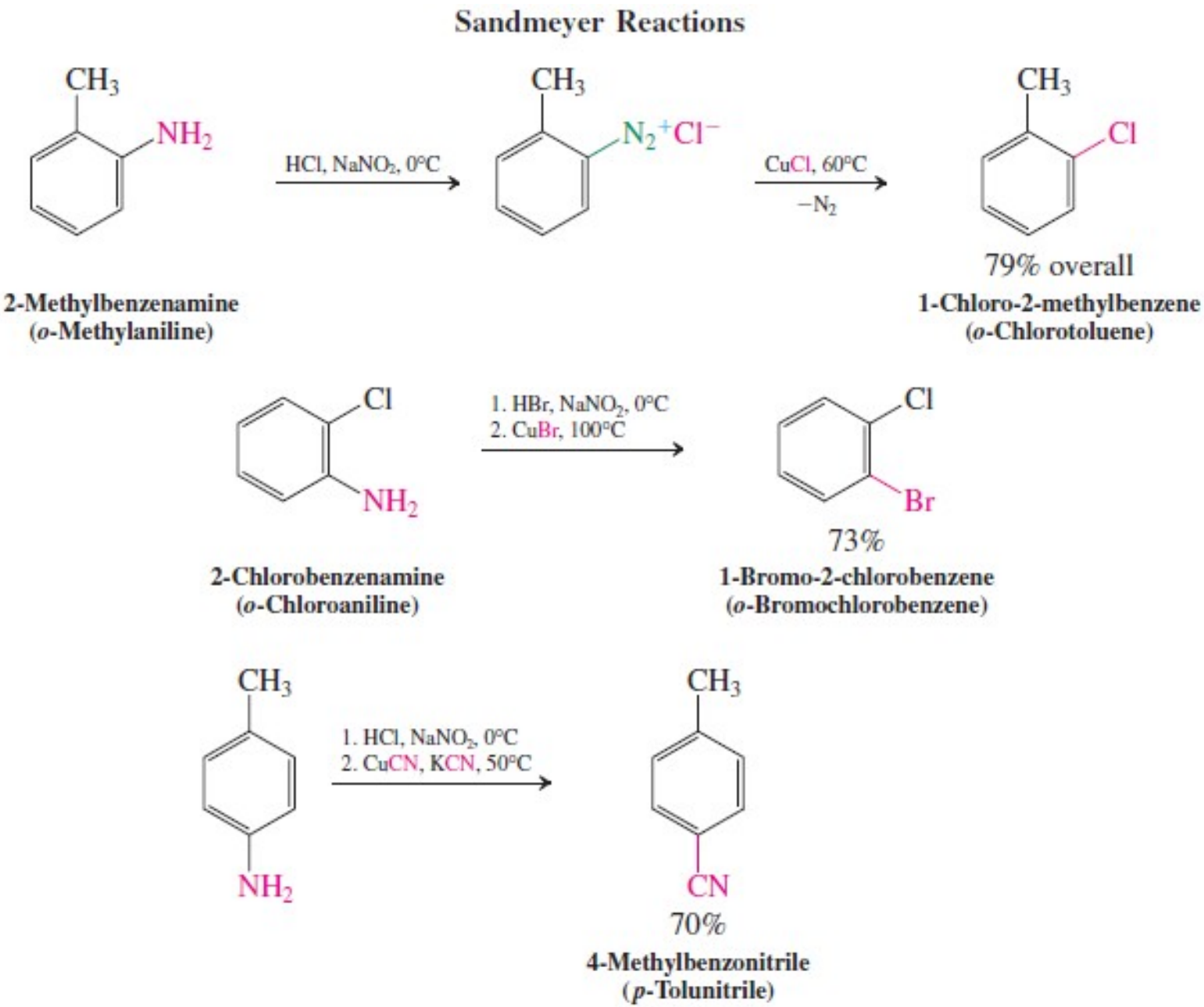
- **The reason for the stability of arenediazonium salts, relative to their alkane counterparts, is resonance and the high energy of the aryl cations formed by loss of nitrogen.**
- **One of the electron pairs making up the aromatic p system can be delocalized into the functional group, which results in charge-separated resonance structures containing a double bond between the benzene ring and the attached nitrogen.**

Resonance in the Benzenediazonium Cation



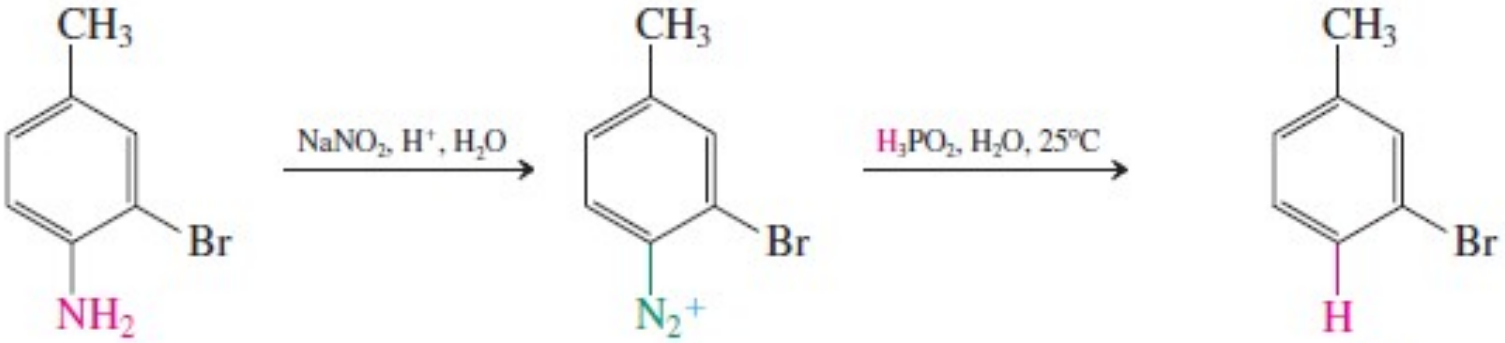


Arenediazonium salts can be converted into other substituted benzenes



- **The diazonium group can be removed by reducing agents. The sequence diazotization –reduction is a way to replace the amino group in arenamines (anilines) with hydrogen.**
- **The reducing agent employed is aqueous hypophosphorous acid, H3PO2. This method is especially useful in syntheses in which an amino group is used as a removable directing substituent in electrophilic aromatic substitution**

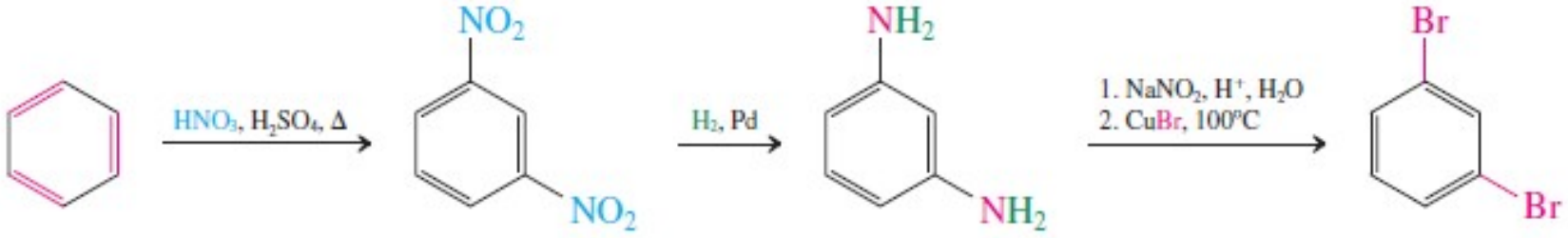
Reductive Removal of a Diazonium Group

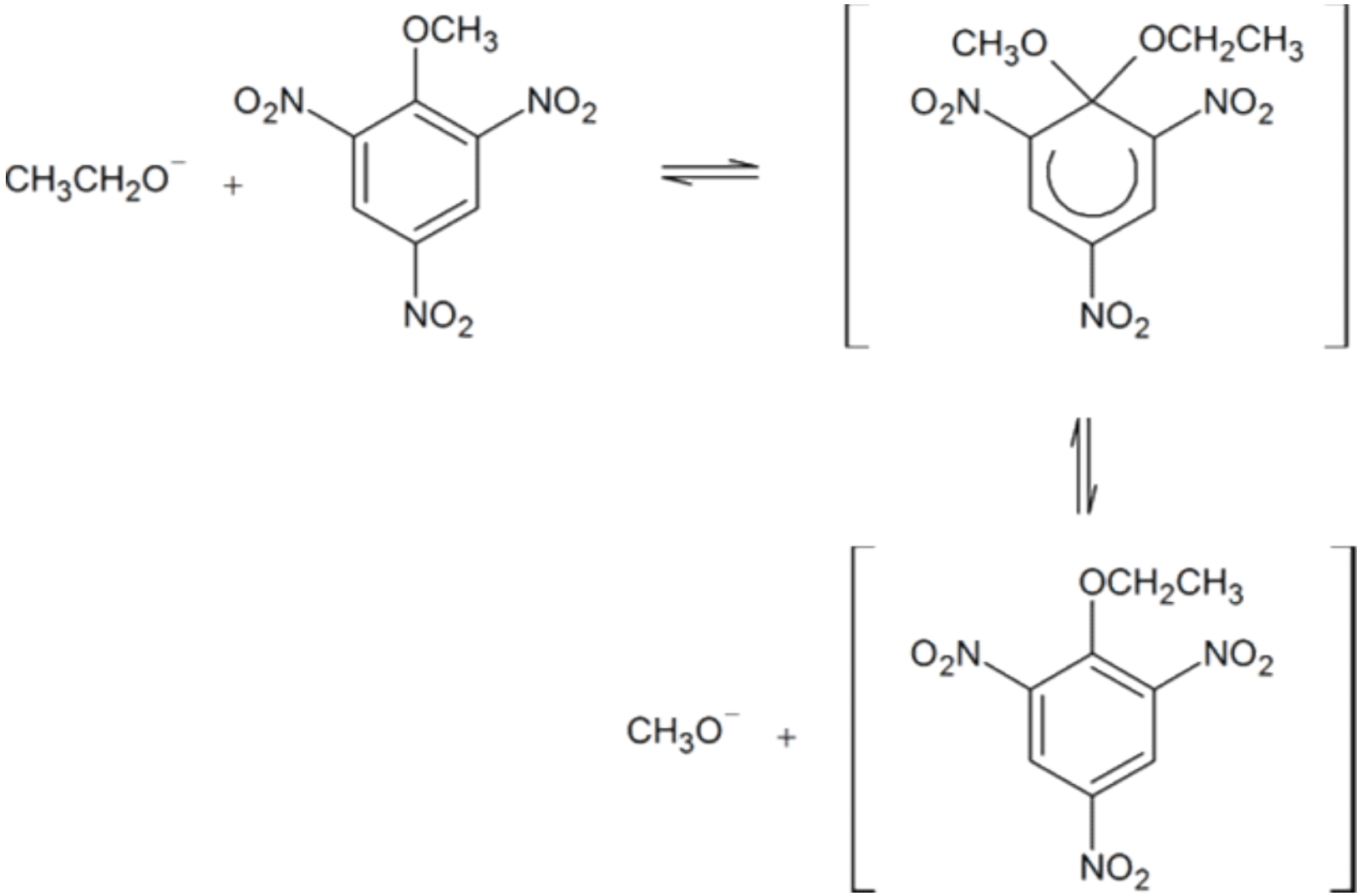


85%

1-Bromo-3-methylbenzene (*m*-Bromotoluene)

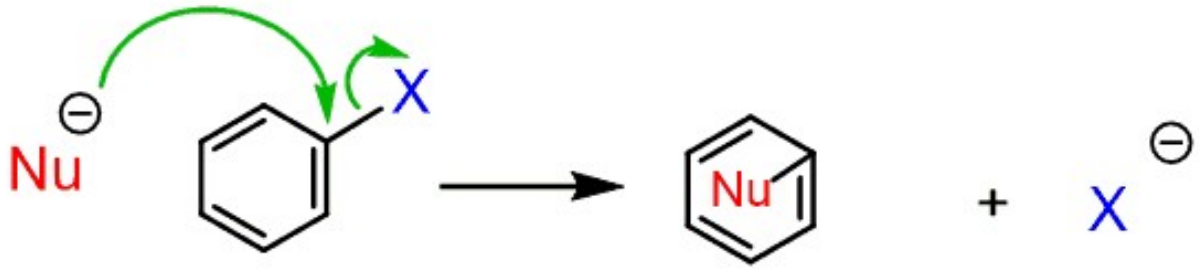
Synthesis of 1,3-Dibromobenzene by Using a Diazotization Strategy




CH3CH2O⁻
+
OCH3
O2N
NO2
NO2
CH3O
OCH2CH3
O2N
NO2
NO2
CH3O⁻
+
OCH2CH3
O2N
NO2
NO2

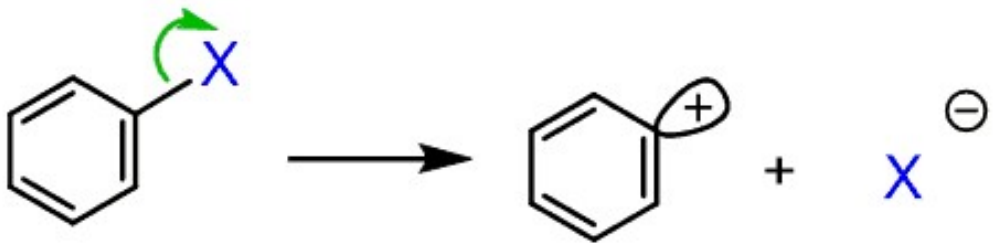
The Mechanism of Nucleophilic Aromatic Substitution

Like in other substitution reactions, the leaving group halide is replaced by a nucleophile. The mechanism of nucleophilic aromatic substitution, however, is different than what we learned in the SN1 and SN2 reactions. The key difference is that aryl halides cannot undergo an SN2 by a backside attack of the nucleophile and, unlike SN1, the loss of the leaving group cannot occur since the phenyl cations are very unstable:



S_N2 - wrong

Impossible structure



S_N1 - wrong

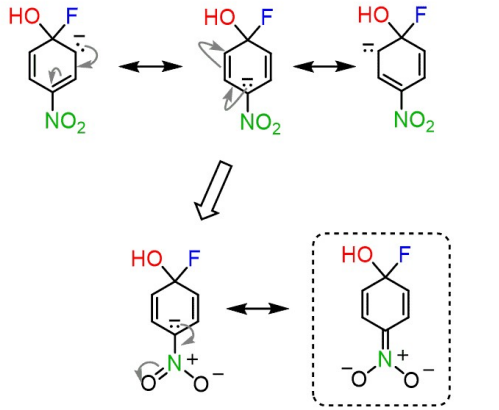
Aryl cation too unstable

Nucleophilic Aromatic Substation

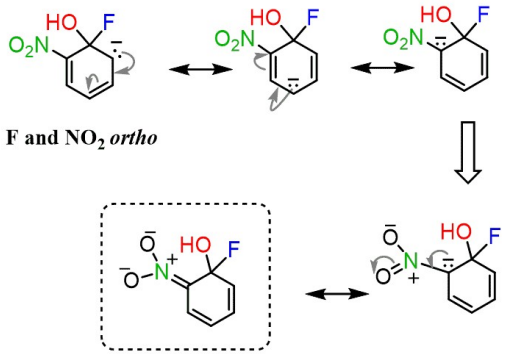
Why must the leaving group be ortho/para to the EWG?

Meisenheimer complex - Resonance stabilization

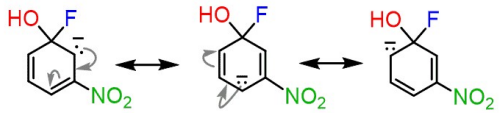
Possibility 1. F and NO_2 are **para**



Possibility 2. F and NO_2 are **ortho**



Possibility 3. F and NO_2 are **meta**



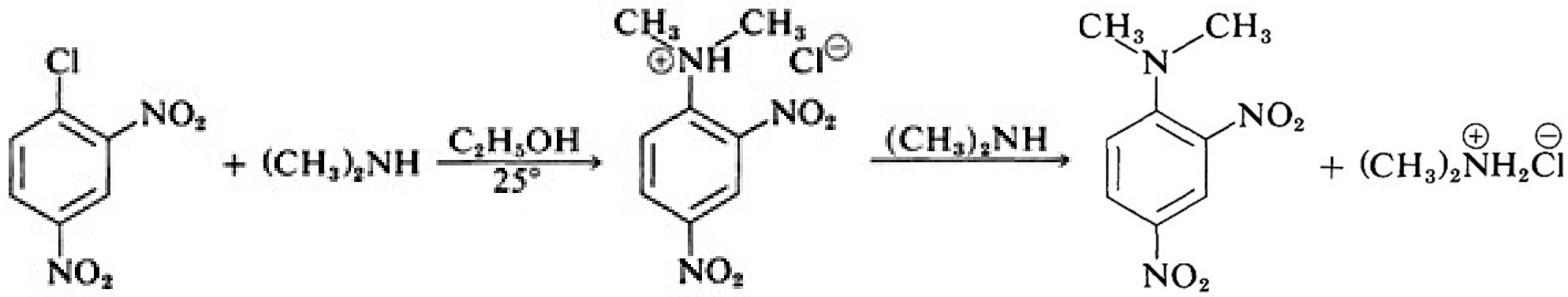
*

NO_2 doesn't get to stabilize the negative charge!

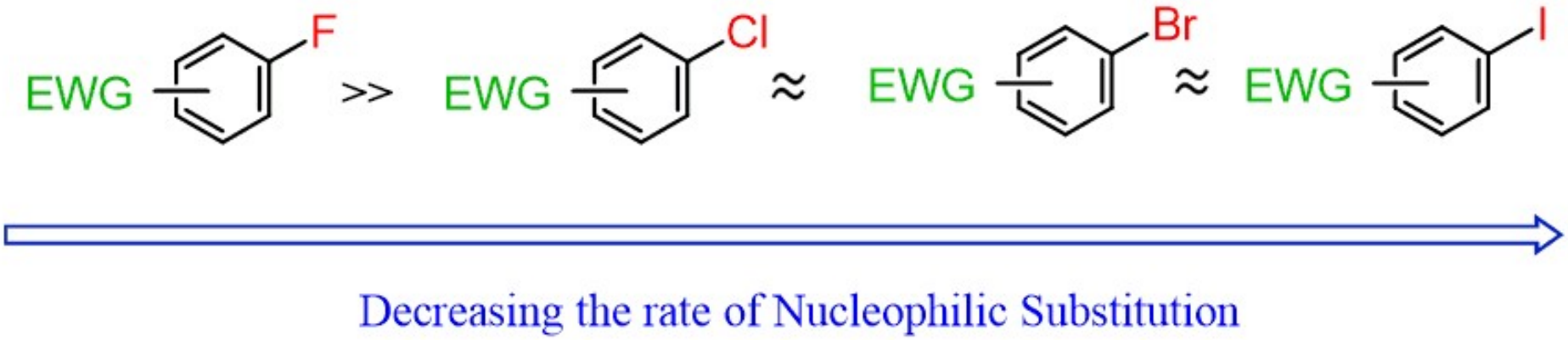
The complex is not as stable

Addition-Elimination Mechanism of Nucleophilic Substitution of Aryl Halides

Although the simple aryl halides are inert to the usual nucleophilic reagents, considerable activation is produced by strongly electron-attracting substituents provided these are located in either the ortho or para positions, or both. For example, the displacement of chloride ion from 1-chloro-2,4-dinitrobenzene by dimethylamine occurs readily in ethanol solution at room temperature. Under the same conditions chlorobenzene completely fails to react; thus the activating influence of the two nitro groups amounts to a factor of at least

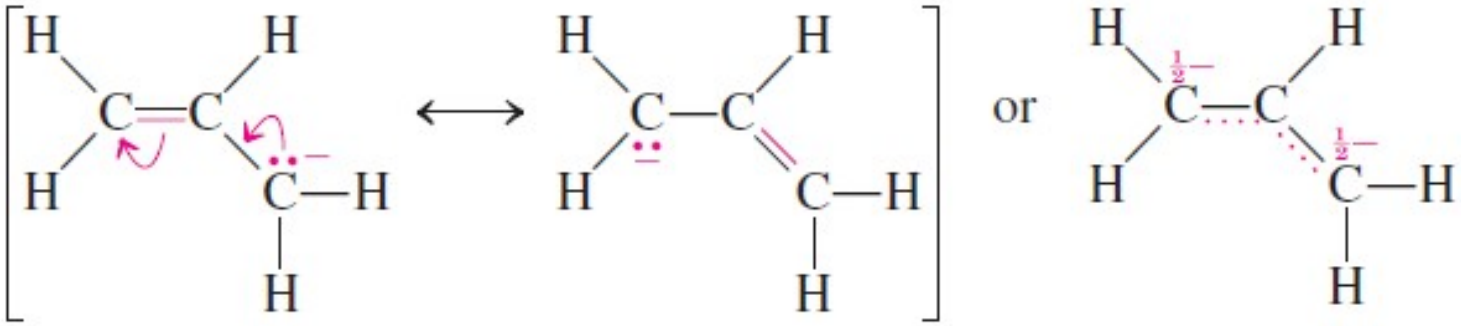


The electronegativity of the halogen increases the rate of the reaction

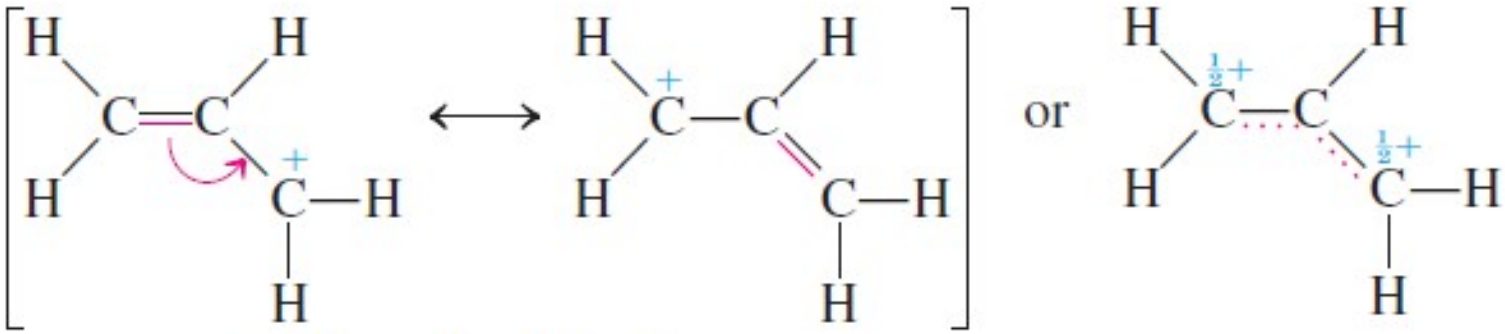


So, the more electronegative the halogen, the better leaving group it is in a nucleophilic aromatic substitution.

And this is the opposite of what we learned in the SN1 and SN2 reactions. We learn that the carbon-fluorine bond is the strongest, and the iodide, being polarizable, was the best leaving group.

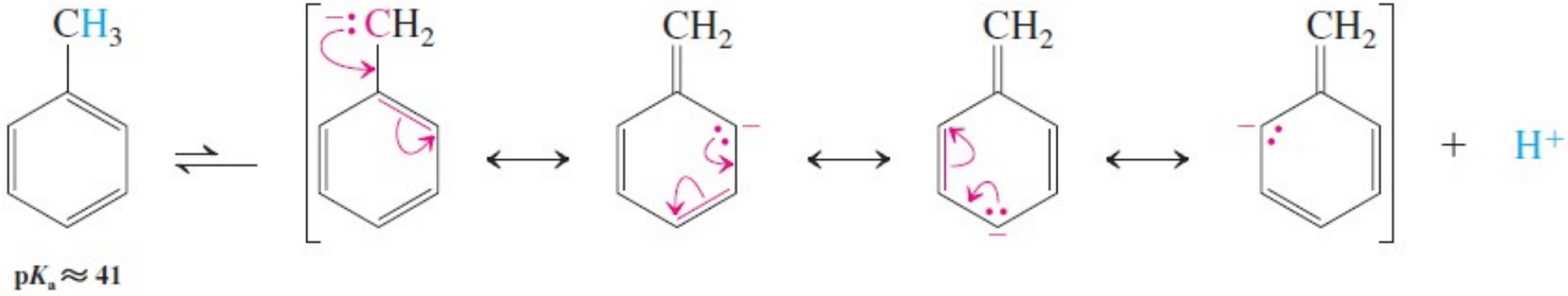


2-Propenyl (allyl) anion

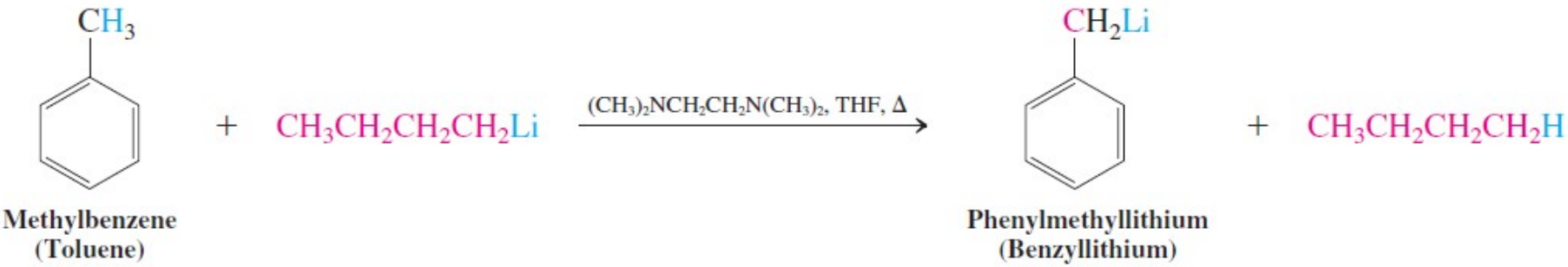


2-Propenyl (allyl) cation

Resonance in Benzylic Anions



Deprotonation of Methylbenzene



Allyl halide>Alkyl halide>>Aryl halide

- **Allyl halides are the most reactive due to resonance stabilization.**
- **Alkyl halides show moderate reactivity depending on their structure and mechanism.**
- **Aryl halides are least reactive under normal nucleophilic substitution conditions due to resonance and bond strength.**